How Deep Is Your Data? Estimating Recursive Generation Depth via Statistical Fingerprints

Anonymous ACL submission

Abstract

As language models increasingly train on web text containing their own outputs, a recursive model→data→model loop arises, yet no method estimates how many such iterations a text has undergone. We formalize *recursive generation depth estimation* as ordinal classification and construct a benchmark of 180K cell-instances (140K unique passages) spanning three model families, three decoding strategies, and four depths under full-replacement chains at 1–1.5B scale. A lightweight pipeline (19 statistical features classified by Random Forest) achieves d1–d3 balanced accuracy of 59–82% across nine model–strategy cells, with Effective Discriminable Depth $\mathrm{EDD}_{0.70} = 3$ in 8/9 cells. Three findings emerge: (1) depth signal concentrates in perplexity distribution tails, corroborated by classifier convergence and independent cross-perplexity features; (2) decoding strategy affects cross-condition transfer more than model architecture, consistent with structurally distinct degradation trajectories; (3) signal attenuates with scale (14B: $\mathrm{EDD}_{0.70}$=2 under nucleus and temperature, 0 under top-k), with reference-model choice as a major confound. Our claims hold within-strategy or when decoding configuration is known; a two-stage routing pipeline recovers +11.5 pp under cross-strategy transfer. Perturbation-based and membership-inference baselines lack ordinal resolution; depth estimation requires distributional statistics beyond binary detection scores.

1 Introduction

The text on which large language models (LLMs) are trained increasingly contains their own outputs. As synthetic content proliferates across the web, a recursive loop emerges: models trained on web corpora ingest text generated by earlier models, and their outputs feed into the next generation's training data (Shumailov et al., 2024; Alemohammad et al., 2024). This model→data→model cycle is no longer hypothetical; it is already occurring in the wild. Yet whether the recursive generation depth of a text can be estimated from its statistical properties remains unstudied, a question that matters because first-generation synthetic text may be benign or even beneficial for training, while deeply recursive text drives the distributional collapse that degrades model quality (Dohmatob et al., 2024). We focus on the 1–1.5B parameter regime under *controlled full-replacement chains*, where each generation fully replaces the prior training set, isolating the depth variable. Real-world contamination involves partial mixing at unknown proportions; full replacement provides an upper bound on depth signal strength (§6).

Existing tools address adjacent but distinct problems. Binary detectors (Hans et al., 2024; Bao et al., 2024; Basani and Chen, 2026) collapse all depths $d \geq 1$ into a single category, a limitation we confirm empirically: their pairwise d1-vs-d2 AUC collapses to chance level (§5.1); watermarking (Kirchenbauer et al., 2023) requires proactive embedding. The model collapse literature (Shumailov et al., 2024; Dohmatob et al., 2024) assumes depth is known a priori; Drayson et al. (2025) filter synthetic data via binary detection without distinguishing depths. No existing method estimates the ordinal recursive generation depth of a text passage from its content alone (Figure 1).

We find that recursive generation leaves quantifiable statistical fingerprints in text, and these fingerprints are **consistent in direction but scale-dependent in magnitude** across the three model families and three decoding strategies studied. Specifically, the perplexity tail feature family (`p90_ppl`, `p75_ppl`, `kurtosis_ppl`) and surprisal variance emerge as dominant depth indicators, with group-level dominance robust across all 9 model–strategy configurations and three importance measures. Our contribution centers on task formalization and systematic empirical characterization; the convergence of four classifiers within 0.8 pp (§5.1) confirms that signal resides in feature design, not classifier architecture. We formalize recursive generation depth estimation as ordinal classification and find that a lightweight pipeline suffices: compute 19 reference-model statistical features (effective dimensionality ≈4–5 due to multicollinearity; §4.2) and classify with a Random Forest (RF) into $K+1$ depth classes. To characterize how recursive generation reshapes these features, we construct the *Depth Degradation Atlas*: a 3×3 grid of model families × decoding strategies spanning 4 depths (180K cell-instances). The atlas reveals strategy-dependent degradation trajectories; we introduce *Effective Discriminable Depth* (EDD) to quantify how far into the recursion chain adjacent depths remain distinguishable.

Our contributions are:

- **Task formalization and metric.** We define recursive generation depth estimation as ordinal classification, introduce the *Effective Discriminable Depth* (EDD) metric, and construct a 180K cell-instance benchmark (3 model families × 3 decoding strategies × 4 depths; 140K unique passages).
- **Depth Degradation Atlas.** The perplexity tail feature family and surprisal variance jointly dominate depth discrimination across all 9 cells; $\mathrm{EDD}_{0.70}$=3 in 8/9 cells (4 under nucleus depth extension), validated by 5-class extension (71–81%); feature group ablation confirms perplexity tail dominance (76.6% alone vs. 78.4% full set).
- **Partial-mixing validation.** Under 25–50% synthetic mixing, $\mathrm{EDD}_{0.70}$=3 is preserved in 5/6 cells with −8–15 pp accuracy attenuation.
- **Generalization boundaries.** Within the tested 1–1.5B regime, cross-model transfer reaches 66.4% and cross-domain (arXiv) 74.1%; at 14B, cross-reference $\mathrm{EDD}_{0.70}$=2; cross-strategy (d1–d3: 45.9%) is a boundary condition, partially recovered by two-stage routing (+11.5 pp).

2 Related Work

Model Collapse Theory. Recursive training on model outputs erodes distributional tails (Shumailov et al., 2024), an effect formalized as exponential tail decay via scaling laws (Dohmatob et al., 2024; Seddik et al., 2024) and extended to autophagous loops in the MAD framework (Alemohammad et al., 2024) and Markovian generation chains (Geng et al., 2026). Bertrand et al. (2024) identify convergence to a fixed point under mixed real-synthetic retraining (grounding our EDD ceiling, §3.5); refinements address generalization-to-memorization transitions (Shi et al., 2025) and broader model classes (Dohmatob et al., 2025). Schaeffer et al. (2025) argue that "model collapse" conflates multiple phenomena; we respond by operationalizing tail-erosion into a deployable ordinal classifier that reveals strategy-dependent trajectories the theory does not predict. All assume depth is known a priori; we address the inverse problem of estimating depth from text alone.

AI Text Detection. The detection literature (Wu et al., 2025) spans statistical (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024; Gehrmann et al., 2019; Verma et al., 2024), geometric (Tulchinskii et al., 2023), and watermarking (Kirchenbauer et al., 2023) methods, with known limits (Sadasivan et al., 2023; Pudasaini et al., 2026). Bevendorff et al. (2025) distinguish attribution from verification; all remain binary, whereas our approach outputs ordinal depth labels along a third axis—recursive iteration.

Membership Inference and Data Provenance. Training data extraction (Carlini et al., 2021) motivates provenance pipelines; Min-K% Prob (Shi et al., 2024) and Datamodels (Ilyas et al., 2022) address membership inference but remain binary; watermarks (Christ et al., 2024) require generator cooperation; TROVE (Zhu et al., 2025) traces sentence-level provenance but targets source attribution. Depth estimation occupies a distinct niche: ordinal, retroactive, and cooperation-free.

Adjacent Tasks. EditLens (Thai et al., 2026) estimates editing extent; Uchendu et al. (2020) classify *which* model generated text; Mohamed et al. (2025) characterize *how* text degrades under iterative generation—none address ordinal depth.

Ordinal Classification. CORAL (Cao et al., 2020) and CLOC (Pitawela et al., 2025) enforce ordinal rank consistency via ordered binary tasks and contrastive losses, respectively; we adopt both as baselines (§5).

Decoding Strategy Effects. Decoding strategies shape text statistics in ways that interact with depth signals (Holtzman et al., 2020; Finlayson et al., 2024; Garcés Arias et al., 2025; Wiher et al., 2022; Basu et al., 2021); our atlas (§4) confirms they induce qualitatively different degradation trajectories.

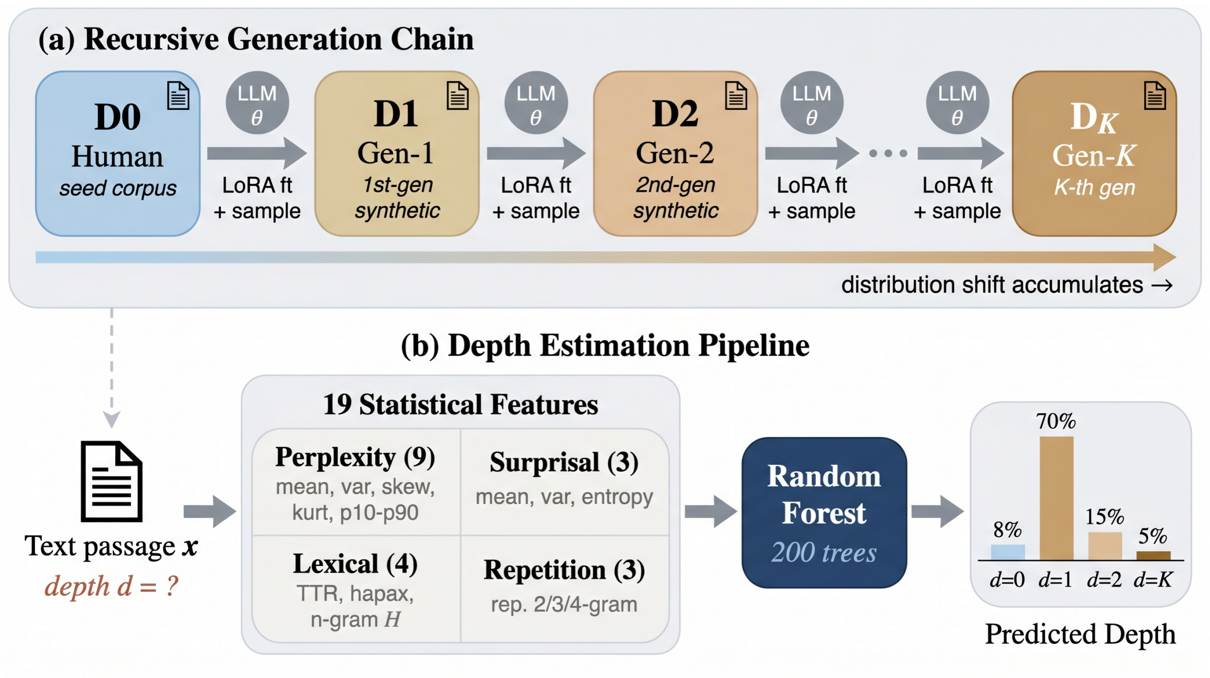

Figure 1: Recursive generation depth estimation. A seed corpus $\mathcal{D}_0$ feeds a LoRA finetuning–sampling loop producing successively deeper corpora. Given a passage x, the task is to estimate its ordinal depth d.

3 Method

3.1 Problem Definition

Let f_θ denote a language model and $\mathcal{D}_0$ a corpus of human-written seed text. We define a *recursive generation chain* of depth K as the sequence $\mathcal{D}_0, \mathcal{D}_1, \ldots, \mathcal{D}_K$, where each $\mathcal{D}_k$ ($k \geq 1$) is produced by finetuning f_θ on $\mathcal{D}_{k-1}$ and sampling from the resulting model. Depth $d{=}0$ denotes human-written text; depth $d{=}k$ ($k \geq 1$) denotes text generated by a model trained on depth-$(k{-}1)$ data.

Given a text passage x, the task is to estimate its recursive generation depth $d \in \{0, 1, \ldots, K\}$. We use *depth estimation* as shorthand throughout; the underlying task is ordinal classification. Depths carry a natural ordering, and $d{=}1$ is structurally closer to $d{=}0$ than $d{=}3$ is, since each generation step compounds distributional drift.

3.2 Multi-Generational Benchmark Construction

We construct a benchmark spanning three axes: model family, decoding strategy, and generation depth.

Model families. We select three 1–1.5B-parameter models with diverse pretraining corpora and architectures: Pythia-1.4B (EleutherAI), OLMo-1B (AI2), and GPT-2 XL (OpenAI, 1.5B). Restricting to a narrow parameter range isolates the effect of architecture and training data from scale.

Decoding strategies. Each model is paired with three decoding strategies: nucleus sampling ($p{=}0.95$) (Holtzman et al., 2020), temperature sampling ($\tau{=}0.9$), and top-k sampling ($k{=}50$). These span the dominant modes of open-ended generation and induce qualitatively different entropy profiles. Our findings on feature importance and discriminability are limited to this grid; transfer to untested models, scales, or strategies is not established.

Depth and scale. The primary benchmark covers depths d0–d3 across all $3 \times 3 = 9$ model–strategy cells, with 5,000 samples per depth per cell (180K cell-instances; 140K unique passages, as d0 samples are drawn from a shared Wikipedia seed corpus across all cells). For depth extension experiments (§4), we extend all three models under nucleus sampling to d0–d5.

Generation pipeline. Starting from a Wikipedia subset ($\mathcal{D}_0$), each cycle: (1) Low-Rank Adaptation (LoRA) finetuning (Hu et al., 2022) on $\mathcal{D}_{k-1}$ ($r{=}16$, $\alpha{=}32$, lr $2{\times}10^{-4}$, effective batch 16, 3 epochs, dropout 0.05; fresh adapter from the base model at each depth; full protocol in Appendix A; cross-seed replication in Appendix B.21); (2) sampling 5,000 passages (max 512 tokens, beginning-of-sequence (BOS) seeded; Appendix B.23) under the specified strategy; (3) $\mathcal{D}_k$ feeds the next cy-

Table 1: The 19 statistical features comprising $\mathbf{f}(x)$, grouped by category. All perplexity and surprisal features are computed from the reference model (Pythia-1.4B).

Category	Features	n
Perplexity statistics	mean, var, skewness, kurtosis, p10, p25, p50, p75, p90	9
Surprisal statistics	mean, var, entropy	3
Lexical diversity	type–token ratio, hapax ratio, bigram entropy, trigram entropy	4
Repetition	repeated 2/3/4-gram fractions	3

cle. Each depth is produced by a model trained on prior-depth data, not by sampling from the unmodified base model. BOS seeding may amplify depth signal; prompted-generation validation (Appendix B.32) confirms signal persistence albeit attenuated (-13.8 pp). No per-passage lineage exists across depths; cross-validation splits do not introduce leakage.

Reference model. All statistical features (§3.3) use Pythia-1.4B as fixed reference model. This overlaps with one generator (Pythia), conferring a 4–11 pp self-reference advantage on that pipeline (Appendix B.17); Pythia results in Table 2 are marked † accordingly. Alternative references yield 67–79%, all exceeding 2.5× chance (Limitations).

3.3 Statistical Feature Design

We extract a feature vector $\mathbf{f}(x)$ comprising 19 candidate features from the reference-model perplexity distribution over each passage x; ablation (§4.2) shows the top 5 achieve 97.6% of full-set performance, and Variance Inflation Factor (VIF) analysis reveals VIF>10 for 14–16 features; effective independent dimensionality is ≈4–5. The features fall into four categories (Table 1).

Perplexity statistics (9 features) capture the per-token perplexity distribution via moments and quantiles (p10–p90); tail quantiles are especially informative as recursive finetuning erodes the high-perplexity tail (§4). *Surprisal statistics* (3 features): mean, variance, and entropy of $-\log p(\text{token})$. *Lexical diversity* (4 features): type–token ratio (TTR), hapax ratio, bigram/trigram entropy. *Repetition* (3 features): repeated n-gram fractions for $n \in \{2, 3, 4\}$.

The design targets the signal predicted by model collapse theory: recursive finetuning progressively erodes high-perplexity tokens (Shumailov et al., 2024), so upper quantiles (p75, p90) and kurtosis track tail erosion monotonically across depths (§4). Surprisal variance captures a complementary effect, dominating at shallow depths where tail erosion is still subtle (§4.2); lexical and repetition features improve cross-strategy robustness despite limited individual discriminative power (Appendix B.28).

3.4 Effective Discriminable Depth

Aggregate multi-class accuracy obscures whether discriminability degrades at specific depth boundaries. We propose the *Effective Discriminable Depth* (EDD) as an operational measure of how far into a recursive chain adjacent depths remain distinguishable.

Pairwise discriminability. Let $\mathcal{D} = \{0, 1, \ldots, K\}$ denote the depth set and $T \in (0.5, 1.0]$ a discrimination threshold. For each adjacent pair $(d, d{+}1)$, we define the pairwise discriminability $\mathrm{AUC}(d, d{+}1)$ as the area under the receiver operating characteristic (ROC) curve of a dedicated binary Random Forest classifier (500 trees, $\lfloor\sqrt{19}\rfloor$ candidate features per split) trained on class-balanced data from $\mathbf{f}(x)$ (§3.3) to distinguish depth-d from depth-$(d{+}1)$ samples, evaluated via 5-fold stratified cross-validation (CV). Each pair uses an independent classifier to avoid decision boundary dilution from multi-class settings.

Definition 1 (Effective Discriminable Depth)**.** The *Effective Discriminable Depth* at threshold T is:

$$\mathrm{EDD}_T = \max\big\{d^*{+}1 \;:\; \forall\, d' \le d^*,\ \mathrm{AUC}(d', d'{+}1) \ge T\big\} \quad (1)$$

where $d^* \in \{0, \ldots, K{-}1\}$. If $\mathrm{AUC}(0, 1) < T$, then $\mathrm{EDD}_T = 0$.

EDD_T reports the longest *contiguous* discriminable prefix: the deepest d such that every pair from $(0, 1)$ through $(d{-}1, d)$ exceeds T. All AUC profiles in the primary benchmark decay monotonically; non-monotonic profiles occur only in cross-scale extensions (Appendix B.18). We adopt $T{=}0.70$ following the "acceptable discrimination" threshold in ROC analysis (Hosmer and Lemeshow, 2000) (8/9 cells achieve $\mathrm{EDD}_{0.70} \ge 3$); sensitivity analysis (§4.3) shows EDD shifts by at most one level across $T \in \{0.65, 0.70, 0.80\}$ in all nine cells.

Classifiers. Our primary classifier is a Random Forest (200 trees, unconstrained depth, 5-fold CV; EDD binary classifiers use 500 trees; seed robustness in Appendix B.29).[1] The RF produces ordinally consistent predictions: 98.2% of errors fall on adjacent depths ($|\Delta| \leq 1$), quadratic weighted kappa (QWK) = 0.89. Comparisons against other classifiers and detection baselines appear in §5.

3.5 Theoretical Motivation

Recursive generation contracts the support of the learned distribution, with low-probability tokens lost first (Shumailov et al., 2024): tail mass $P_k(\text{ppl} > t)$ decreases monotonically in depth k. This tail erosion predicts the perplexity tail feature family as a depth marker, confirmed jointly with surprisal variance across all 9 cells (§4.2). Surprisal variance captures complementary distributional narrowing: as the support contracts, per-token surprise becomes more predictable.

EDD upper bound. Tail mass at depth k scales as γ^k for a contraction factor $\gamma \in (0, 1)$ (Dohmatob et al., 2024); EDD reaches its upper bound when adjacent-depth feature distance falls below classifier resolution. Our empirical AUC-excess contraction (ratio 0.80 at d1v2→d2v3, 0.69 at d3v4→d4v5; §4.3) is consistent with geometric decay approaching a fixed point (Bertrand et al., 2024). 5-class extensions to d4 (71–81%, chance = 20%; Appendix B.22) confirm depth 4 discriminability; mean perplexity saturates before tail statistics, placing a physical ceiling on EDD.

Scale-dependent attenuation. Larger models approximate the training distribution more faithfully per step (γ closer to 1), predicting weaker per-step signal. Qwen2.5-14B supports this: d0-vs-d1 AUC drops to 0.643 (vs. >0.98 at 1–1.5B; §5.4); 7–8B results are mixed, and additional confounds likely interact with the scale effect (Appendix B.18).

Strategy-dependent convergence. The contraction rate γ varies with decoding strategy. Temperature scaling ($\tau < 1$) sharpens the logit distribution uniformly and accelerates convergence to the mode; temperature depth-d text should therefore resemble nucleus depth-$(d+1)$ text in tail statistics. GPT-2 XL confirms this: temperature d1 median `p90_ppl` (92.2) matches the nucleus d2 median (92.3). Top-k truncation removes the low-probability tokens that carry depth signal, predicting weaker discriminability; top-k cells yield the lowest accuracy (68.9–75.8%) across the atlas (§4.4).

[1]Statistical tests are not corrected for multiple comparisons; p-values should be interpreted as exploratory.

Table 2: Four-class RF accuracy (%) with bootstrap 95% confidence interval (CI; ±half-width; B=10,000). RF with 200 trees, evaluated via 5-fold stratified CV (seed=42). Parenthesized: d1–d3 balanced accuracy from the same 4-class RF (chance = 25% / 33%). Depth-0 recall exceeds 97% in all cells. Within-distribution ordinal quality: mean absolute error (MAE) = 0.23, QWK = 0.89 ($\text{QWK}_{\text{d1–d3}}$ = 0.74); 98.2% of errors fall on adjacent depths ($|\Delta| \leq 1$). †Self-reference scoring (Pythia-1.4B is both generator and reference model); 4–11 pp advantage over cross-model references (Appendix B.17). Cross-cell comparisons involving Pythia should be interpreted with this caveat.

	Nucleus	Temp. τ=0.9	Top-k
Pythia-1.4B†	78.7±0.5 (71.8)	76.8±0.5 (69.3)	68.9±0.6 (58.9)
OLMo-1B	78.9±0.5 (72.7)	82.3±0.5 (76.6)	74.4±0.5 (66.2)
GPT-2 XL	**84.3**±0.5 (79.6)	**85.9**±0.5 (81.5)	**75.8**±0.6 (68.1)

4 The Depth Degradation Atlas

We present the Depth Degradation Atlas: a systematic empirical map of how statistical fingerprints evolve across recursive generation depths, models, and decoding strategies. The atlas spans $3 \times 3 = 9$ model–strategy cells at the 1–1.5B parameter scale, evaluated at depths $d \in \{0, 1, 2, 3\}$, with depth extensions to d=5 for cross-model EDD validation (§4.3).

4.1 In-Distribution Performance

Table 2 reports four-class Random Forest accuracy (5-fold CV, 200 trees) across all nine cells. Accuracy ranges from 68.9% (Pythia, top-k) to 85.9% (GPT-2 XL, temperature) four-class (d1–d3 balanced: 58.9–81.5%), all exceeding the 25%/33% chance baselines. Two axis effects emerge. GPT-2 XL achieves the highest accuracy under every strategy (75.8–85.9%), followed by OLMo (74.4–82.3%) and Pythia (68.9–78.7%). Top-k sampling yields the lowest accuracy in every model. The mean within-model strategy gap is 9.3 pp, comparable to the 7.2 pp mean within-strategy model gap; decoding strategy contributes at least as much as model family to performance variance.

Per-class recall. Depth-0 recall exceeds 97% in all 9 cells. Among generated depths, d2 consistently yields the lowest recall (44–73%), reflecting its position at the center of the depth gradient where features overlap with both d1 and d3, consistent with d2-vs-d3 as the EDD bottleneck (§4.3; full per-class recall and confusion matrices in Appendix B.5, B.6).

4.2 Feature Consistency Across the Grid

Multicollinearity is substantial (VIF>10 for 14–16 of 19 features); importance rankings below indicate feature *family* contributions rather than individual effects; permutation importance (Appendix B.2) partially mitigates collinearity bias (Strobl et al., 2008; Hooker et al., 2021). Despite the 17.0 pp accuracy spread across cells, the perplexity tail statistics family (`p90_ppl`, `p75_ppl`, `kurtosis_ppl`) and surprisal variance (`var_surprisal`) dominate depth discrimination in every setting. Within this family, `p90_ppl` and `var_surprisal` are consistently prominent, though individual rankings should be interpreted cautiously given VIF>10; group-level ablation (§4.2, below) provides the collinearity-robust confirmation.[2] Conditional permutation importance (Strobl et al., 2008) confirms `var_surprisal` and `p90_ppl` as top-2 in 8/9 cells (bootstrap rank 1.4 and 1.7), consistent with Gini rankings (Appendix B.3). Full per-cell rankings appear in Table 7 (Appendix B.4); Figure 2 visualizes the importance landscape across all nine cells.

Cross-model feature-rank Spearman correlation averages $\bar{\rho} = 0.868$ ($[0.619, 0.954]$): models agree on which features carry depth information despite differing in absolute accuracy. Top-k yields the highest cross-model rank correlation ($\rho = 0.811$), as vocabulary truncation imposes a more uniform feature landscape at the cost of weaker signal (§4.4).

Feature ablation confirms this concentration: the top-5 features achieve 97.6% of full accuracy (76.5% vs. 78.4%); in-distribution group-level ablation shows perplexity features alone reach 76.6%, surprisal 73.2%, lexical diversity ≤47%, and repetition ≤44% (Appendix B.28).[3]

The feature hierarchy shifts with depth. At shallow boundaries (d1 vs. d2), `var_surprisal` dominates importance; at deeper boundaries (d3 vs. d4, d4 vs. d5), tail perplexity statistics (`p90_ppl`, `kurtosis_ppl`) take over. This transition aligns with model-collapse dynamics (Shumailov et al., 2024): early recursion narrows the surprisal distribution (captured by its variance), while later steps erode the perplexity tail (captured by upper quantiles and kurtosis).

[2]The group-level dominance of perplexity tail statistics is robust across importance measures (Gini, SHapley Additive exPlanations (SHAP), permutation; Kendall's W=0.90, p<0.005).

[3]Best single feature (`p90_ppl`) achieves 58.3%, providing a more competitive single-feature baseline than mean perplexity (25.1%).

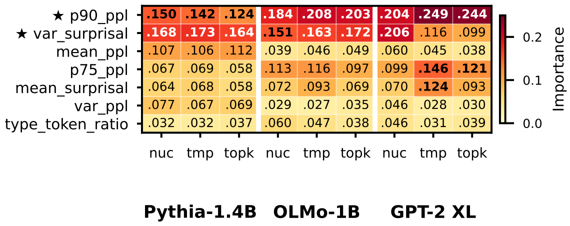

Figure 2: Atlas feature importance heatmap for the top-7 features. Rows: features ranked by mean Gini importance; columns: 9 model–strategy cells. Stars mark the two universal top-2 features (`p90_ppl`, `var_surprisal`). Feature group ablation in Appendix B.28.

4.3 EDD Analysis

Depth features exhibit strong monotonic trends (Spearman $\rho = -1.0$ for perplexity tail statistics; Figure 3); inter-depth distances are non-uniform (d0-vs-d1 AUC >0.98 in all 9 cells, d2-vs-d3: 0.69–0.88).

We extend the benchmark to depth d=5 for all three models under nucleus sampling. Each model's depth-3 LoRA checkpoint is used to fine-tune and generate depth-4 data, and the process repeats for depth 5; feature extraction uses the same reference model (Pythia-1.4B). We train dedicated binary RF classifiers (500 trees) for each adjacent pair, as defined in Eq. 1.

AUC decay and cross-model convergence. For Pythia-1.4B (nucleus), AUC decays monotonically from 0.999 (d0v1) to 0.683 (d4v5); AUC-excess contraction ratios (0.80 at d1v2→d2v3, 0.69 at d3v4→d4v5) show convergence toward indistinguishability accelerates with depth (Figure 4). The d3-vs-d4 AUC clusters near T=0.75 for all three models (Pythia 0.784, OLMo 0.765, GPT-2 XL 0.750), while d4-vs-d5 converges to a narrow band (0.681–0.684, range 0.003), consistent with an architecture-independent depth ceiling within the 1–1.5B nucleus setting. Bootstrap testing (H_0: AUC $\leq$ 0.75; Bonferroni-corrected for 3 comparisons, α=0.0167) yields p = 0.001 (Pythia), 0.091 (OLMo), 0.505 (GPT-2 XL).

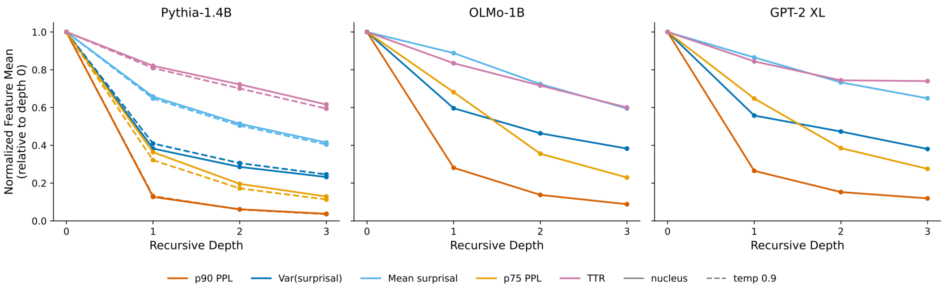

Figure 3: Feature trajectories (normalized to depth 0) across models and strategies. The $d{=}0 \rightarrow d{=}1$ gap dominates; tail features (`p90_ppl`, `p75_ppl`) decline most steeply, consistent with tail erosion (§3.5).

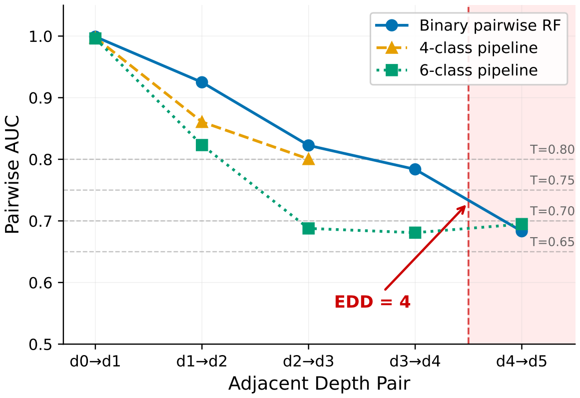

Figure 4: AUC decay curves for Pythia-1.4B (nucleus, depths 0–5) under three pipeline configurations: binary pairwise RF, 4-class RF, and 6-class RF. Horizontal dashed lines mark $T \in \{0.65, 0.70, 0.75, 0.80\}$. All curves converge below 0.70 at d4-vs-d5. Shaded bands: bootstrap 95% CIs.

EDD threshold sensitivity. Across the 9-cell benchmark (depths 0–3), all cells achieve EDD=3 at $T{=}0.65$; at $T{=}0.70$, 8/9 cells; at $T{=}0.75$, 7/9; at $T{=}0.80$, 5/9. Top-k configurations lose EDD=3 first (Pythia top-k at $T{=}0.70$, OLMo top-k at $T{=}0.75$); d2-vs-d3 is the consistent bottleneck.

4.4 Strategy-Dependent Trajectories

The three strategies induce qualitatively distinct degradation trajectories. Temperature ($\tau{=}0.9$) yields the highest accuracy in 2/3 models (GPT-2 XL 85.9%, OLMo 82.3%) because uniform logit scaling accelerates convergence to an attractor (Shumailov et al., 2024), amplifying the fingerprint but creating a boundary reversal risk under cross-strategy transfer (§5; temperature robustness analysis in Appendix B.11). Top-k ($k{=}50$) produces the lowest accuracy (68.9–75.8%) because hard vocabulary truncation removes depth-discriminative low-probability tokens. The strategy effect (9.3 pp range) exceeds the model effect (7.2 pp); deployment-time knowledge of the decoding strategy matters at least as much as the generator model.

5 Generalization and Robustness

We test whether atlas accuracy (68.9–85.9% four-class; d1–d3: 58.9–81.5%) reflects genuine statistical regularities across eight generalization axes (five below, three in Appendix).

5.1 Classifier Comparison and Surface-Text Ablation

Table 3 compares methods across all 9 cells. Four classifiers (RF, XGBoost, ordinal multilayer perceptron (MLP), prototypical) converge within 0.8 pp under cross-model transfer (Appendix B.25), which confirms a feature-determined accuracy ceiling.

Neural end-to-end baselines. DeBERTa-v3-base (LoRA, 53–61%) and RoBERTa-DCOL (ordinal contrastive, 64.5%) lag RF by 14–33 pp; note that RF features access reference-model log-probabilities unavailable to end-to-end pretrained language model (PLM) baselines (§6).

Binary detectors and membership inference. Perturbation-based (DetectGPT, 28.8%) and membership-inference (Min-K% Prob, 33.2%) methods produce scores with no ordinal resolution (Table 3); Binoculars d1v2 AUC = 0.50–0.53 vs. RF 0.93 (Appendix B.30). Binoculars convergence with our pipeline is discussed in §6.

Surface-text ablation. Single-feature regression on mean perplexity (25.1%) and zero-shot LLMs (best 36.2%; Appendix B.10) confirm depth signal is inaccessible from surface text.

5.2 Cross-Model Transfer

Mean transfer accuracy is 66.4% (2.7× chance; Table 4; d1–d3: 55.4%); d2 worst case 26.8% (Pythia holdout; confusion matrices in Figure 5).

Table 3: Method comparison (RF mean across 9 cells, 5-fold CV). Chance = 25% / 33.3% (d1–d3). †Adapted from binary detection; Binoculars 11-feat: p>0.05 vs. 19-feat RF (paired permutation test).

Method	Calibration	4-Cl (%)	d1–d3 (%)
RF 19-feature (ours)	RF	78.4	**71.6**
Binoculars† (Hans et al., 2024)	RF (11 feat)	78.1	—
Merged (30 feat)†	RF	**82.6**	—
Binoculars (Hans et al., 2024)	threshold	69.8	—
Binoculars (Hans et al., 2024)	RF (1 feat)	63.7	—
RoBERTa-DCOL (ordinal)	contrastive	64.5	—
DeBERTa-v3 (end-to-end)	cross-entropy	53.1–60.6	—
Zero-shot LLM (best)	—	36.2	—
Min-K% Prob (Shi et al., 2024)	CORAL	33.2	40.4
DetectGPT (Mitchell et al., 2023)	best-of-3	28.8	24.3
Random	—	25.0	33.3

Table 4: Cross-model transfer (leave-one-out, nucleus only). Chance = 25% / 33% (d1–d3). d1–d3 mean 55.4%; QWK = 0.81; 94.8% errors at $|\Delta| \leq 1$.

Holdout	Acc (%)	d0 Rec.	d1 Rec.	d2 Rec.	d3 Rec.
Pythia	62.4	100.0	49.3	26.8	73.6
OLMo	**69.2**	99.3	75.5	43.3	58.8
GPT-2 XL	67.6	98.5	74.7	52.6	44.5
Mean	66.4	99.3	66.5	40.9	59.0

5.3 Cross-Strategy Transfer

Cross-strategy transfer reveals a structural barrier: d1–d3 balanced accuracy drops to 45.9% (four-class: 59.1%), 7 pp below cross-model transfer (Table 9; full per-model results and Figure 6 in Appendix B.7). A two-stage pipeline (strategy classifier → depth classifier) recovers much of this gap (70.6%, +11.5 pp; oracle: 78.4%; Appendix B.33). Our depth estimation claims hold most strongly within-strategy or when the decoding configuration is known.

5.4 Preliminary Cross-Scale Observations (14B)

At 14B (Qwen2.5-14B), self-reference yields 50.8% (2× chance, $\mathrm{EDD}_{0.70}$=0); switching to cross-reference (Pythia-1.4B) shifts accuracy by −6.8 pp to 44.1% but raises $\mathrm{EDD}_{0.70}$ to 2 for nucleus and temperature; reference-model choice is therefore a major confound in cross-scale comparisons (Appendix B.18). At matched 1.5B scale, instruction tuning preserves depth signal (Appendix B.26). Cross-domain transfer to arXiv abstracts reaches 74.1% (Appendix B.14).

Partial-mixing (accumulation). Real-world contamination mixes synthetic and human text rather than fully replacing the training set. We test robustness under 25% and 50% synthetic mixing across three model–strategy cells (Table 10). At 25% mixing, all three cells preserve $\mathrm{EDD}_{0.70}$=3 with 8–15 pp accuracy attenuation relative to full replacement. At 50% mixing, 2/3 cells retain EDD=3; the sole failure (GPT-2 XL, d2v3 AUC = 0.695) falls 0.5 pp below threshold.

6 Discussion and Conclusion

Four importance measures converge on the same ranking (Kendall's W=0.90); Binoculars cross-perplexity features independently reach 78.1% (merged: 82.6%; Appendix B.31); four classifiers converge within 0.8 pp (§5.1); CORAL adds +0.3 pp (Appendix B.25). The accuracy ceiling thus resides in the feature space, not the classifier. End-to-end classifiers lag by 14–33 pp (d2 F1 ≤ 0.26; Appendix B.13, B.9), likely because depth signal resides in passage-level distributional tail statistics rather than local coherence patterns. The strategy effect (9.3 pp) exceeds the model effect (7.2 pp), so deployment requires conditioning on decoding configuration (§5.3); the fixed reference model degrades by 4–11 pp under cross-architecture alternatives (Appendix B.17), and a 7B self-reference probe reveals a scale-dependent ceiling ($\mathrm{EDD}_{0.70}$=0; Appendix B.19). Accumulation experiments and cross-boundary chains (84.8%; Appendix B.8, B.20) indicate that depth fingerprints are process-level rather than model-specific.

Practical implications. Ordinal depth labels enable tiered data curation that binary detection cannot support: a pipeline can retain first-generation text (d=1), which may still carry training value, while filtering d≥2 content where collapse compounds. Since binary detectors collapse all d≥1 with chance-level pairwise AUC between adjacent depths (§5.1), they cannot implement such depth-aware policies. The lightweight feature extraction (19 statistics from a single reference-model forward pass) makes inline deployment practical for provenance auditing.

In summary, recursive generation leaves quantifiable statistical fingerprints that a lightweight pipeline resolves into ordinal depth classes ($\mathrm{EDD}_{0.70}$=3 in 8/9 cells), enabling more targeted data filtering than binary detection for provenance auditing and collapse monitoring. Open questions include reference-model alignment beyond 1.5B (§5.4), paraphrase robustness, finer-grained mixing (§5.4), and extension to instruction-tuned models.

Limitations

Our benchmark uses full-replacement chains at 1–1.5B parameters; real-world contamination involves accumulation at unknown mixing ratios (per-passage robustness in Appendix B.24), and depth signal attenuates at larger scales (§5.4), with reference-model mismatch contributing to the observed degradation (Appendix B.18). Cross-strategy transfer can invert adjacent-depth predictions, requiring strategy-aware deployment (§5.3). On held-out web-crawl corpora, pooled-classifier raw false-positive rates range from 15% to 34%, reducible to $\leq$7% via posterior-threshold calibration (Appendix B.15); robustness to post-hoc paraphrasing remains untested.

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A LoRA Fine-Tuning Protocol

Table 5 lists the full LoRA fine-tuning configuration used for recursive chain construction (§3.2). At each depth k, a fresh LoRA adapter is initialized on the base model and trained on $\mathcal{D}_{k-1}$; adapters are not inherited across depths. The HuggingFace Trainer default linear decay schedule is used with no warmup. Target modules are model-specific: `query_key_value` (Pythia), `att_proj` (OLMo), and `c_attn` (GPT-2 XL). Cross-scale 7–8B experiments (Appendix B.18) use Quantized LoRA (QLoRA; 4-bit NF4) with r=16, α=32; the 14B model uses FP16 LoRA with the same rank and scaling.

Table 5: LoRA fine-tuning hyperparameters for the primary 1–1.5B benchmark. Identical across all three model families and all depth levels.

Parameter	Value
LoRA rank r	16
LoRA alpha α	32
LoRA dropout	0.05
Bias	none
Epochs per depth	3
Learning rate	2×10^{-4}
LR schedule	linear decay (no warmup)
Batch size (effective)	16 (8 × 2 grad. accum.)
Max sequence length	512 tokens
Precision	bfloat16
Adapter init	from scratch (base model)
Seed	42 (primary), 43 (replication)

B Additional Results

B.1 Top-k Feature Trajectories

Feature trajectories for nucleus and temperature sampling appear in Figure 3 (§4.3). Top-k (k=50) trajectories follow the same monotonically decreasing trend for all features but with compressed inter-depth spacing, consistent with the signal attenuation effect discussed in §4.4. Concretely, the d0→d1 drop in `p90_ppl` under top-k is 78% (vs. 87% for nucleus), and the d2→d3 reduction is 31% (vs. 39% for nucleus). Despite the magnitude reduction, feature rank ordering across depths is preserved, which explains why ordinal classifiers maintain directionality even under top-k conditions.

B.2 Permutation Importance

To verify that the Gini-based importance rankings in §4.2 are not an artifact of the impurity measure, we compute permutation importance (5-fold CV, 10 repeats) for all 9 atlas cells. The top-2 features are stable across measures: `p90_ppl` ranks in the top 2 in 9/9 cells under both Gini and permutation importance, and `var_surprisal` in 8/9 (vs. 7/9 for Gini, with the additional cell being OLMo top-k; the single SHAP exception is GPT-2 XL top-k, where `p75_ppl` displaces `var_surprisal` to rank 3). Kendall's W=0.90 across all three methods (Gini, permutation, SHAP) indicates strong concordance (p<0.005). Pairwise Spearman correlations are: SHAP–Gini ρ=0.93, SHAP–permutation ρ=0.84, Gini–permutation ρ=0.76. Per-cell top-5 overlap between SHAP and permutation importance is 3–4 out of 5 features; the aggregated (mean-across-cells) top-5 sets are identical; the top features are therefore robust to the choice of importance measure at the aggregate level, though per-cell rankings exhibit model-dependent variation.

B.3 Conditional Permutation Importance under Multicollinearity

With 14–16 of 19 features exhibiting VIF>10 (§4.2), standard permutation importance may inflate contributions of correlated features: permuting one member of a collinear set allows others to compensate, masking true individual effects. We compute conditional permutation importance (CPI) (Strobl et al., 2008), which permutes each feature *within strata defined by its correlated partners*, isolating the unique contribution after controlling for multicollinearity.

Setup. CPI is computed on all 9 model×strategy cells. For each cell, 1,000 bootstrap resamples produce a rank distribution per feature; the 95% bootstrap CI width quantifies rank stability (width $\leq$ 1.0 indicates near-invariant rankings across resamples).

Results. Table 6 summarizes the top-5 features by mean CPI across cells. `var_surprisal` (avg rank 1.4, CPI = 0.260) and `p90_ppl` (avg rank 1.7, CPI = 0.238) are the top-2 in 8/9 cells, with bootstrap rank CI widths of 0.44 and 0.22—both $\leq$ 0.5, indicating near-invariant rankings across resamples. Ranks 3–5 are `p75_ppl` (CPI = 0.063, CI width = 0.78), `p50_ppl` (0.016, width = 2.67), and a cell-dependent set (`p25_ppl` or `mean_ppl`); rank stability decreases beyond the top-2, with CI widths of 2.7–3.7 for ranks 4–5.

Two exceptions exist among GPT-2 XL cells: in temperature sampling, `p90_ppl`

Table 6: Conditional permutation importance aggregated across 9 model×strategy cells (1,000 bootstrap resamples). Avg Rank and Avg CPI: means over cells. Rank CI Width: mean 95% bootstrap CI width for the rank. Bottom rows: feature-family averages.

Feature / Family	Avg Rank	Avg CPI	Rank CI Width
`var_surprisal`	1.4	0.260	0.44
`p90_ppl`	1.7	0.238	0.22
`p75_ppl`	3.1	0.063	0.78
`p50_ppl`	5.0	0.016	2.67
`p25_ppl`†	5.8	0.012	3.67
`mean_ppl`†	6.3	0.008	3.44
Surprisal family	7.3	0.089	—
Perplexity family	7.5	0.039	—
Lexical family	9.7	0.004	—

†Rank 5–6 varies by cell; `trigram_entropy` appears in 1 cell.

(CPI = 0.419) exceeds `var_surprisal` (0.093) while both remain in the top-2; in top-k sampling, `p90_ppl` (0.475) and `p75_ppl` (0.127) push `var_surprisal` to rank 3 (0.037), the only cell where it exits the top-2. Both reversals are consistent with the Gini rankings for these cells (Table 7).

At the family level, surprisal (avg CPI = 0.089, rank = 7.3) and perplexity (0.039, rank = 7.5) are closely matched, while lexical features (0.004, rank = 9.7) contribute minimally—consistent with the group ablation in §4.2.

Consistency with Gini rankings. CPI and Gini rankings agree on the dominant feature (`var_surprisal` or `p90_ppl`) in 8/9 cells; the sole disagreement (GPT-2 XL top-k) involves a swap between rank-2 and rank-3 features. Bootstrap rank CI widths ≤ 0.5 for the top-2 features confirm that these rankings are not artifacts of collinearity or resampling variability. Controlling for multicollinearity via conditional permutation corroborates the Gini- and SHAP-based importance hierarchy reported in the main text.

B.4 Per-Cell Gini Feature Rankings

Table 7: Top-5 features by Gini importance. **Bold**: top-2. `p90_ppl` top-2 in 9/9; `var_surprisal` 8/9 under SHAP (7/9 under Gini). Individual rankings are approximate due to multicollinearity (VIF>10 for 14–16/19 features).

Cell	Top-5 features (Gini importance, ranked)
Pythia, nuc.	**var_surprisal (.17)**, **p90_ppl (.15)**, mean_ppl (.11), var_ppl (.08), p75_ppl (.07)
Pythia, τ	**var_surprisal (.17)**, **p90_ppl (.14)**, mean_ppl (.11), p75_ppl (.07), mean_surprisal (.07)
Pythia, top-k	**var_surprisal (.16)**, **p90_ppl (.12)**, mean_ppl (.11), var_ppl (.07), mean_surprisal (.06)
GPT-2 XL, nuc.	**var_surprisal (.21)**, **p90_ppl (.21)**, p75_ppl (.10), mean_surprisal (.07), mean_ppl (.06)
GPT-2 XL, τ	**p90_ppl (.25)**, **p75_ppl (.15)**, mean_surprisal (.13), var_surprisal (.12), p50_ppl (.05)
GPT-2 XL, top-k	**p90_ppl (.24)**, **p75_ppl (.12)**, var_surprisal (.10), mean_surprisal (.09), type_token_ratio (.04)
OLMo, nuc.	**p90_ppl (.18)**, **var_surprisal (.15)**, p75_ppl (.11), mean_surprisal (.07), type_token_ratio (.06)
OLMo, τ	**p90_ppl (.21)**, **var_surprisal (.16)**, p75_ppl (.12), mean_surprisal (.09), type_token_ratio (.05)
OLMo, top-k	**p90_ppl (.20)**, **var_surprisal (.17)**, p75_ppl (.10), mean_surprisal (.07), mean_ppl (.05)

B.5 Per-Class Recall

Table 8 reports per-class recall across all 9 atlas cells (4-class in-distribution, 5-fold CV). Depth 0 recall exceeds 97% uniformly. Among generated depths, d2 is the hardest class in every cell; the deficit is most pronounced under top-k sampling (46–53%), where vocabulary truncation removes the tail tokens that carry depth-discriminative signal.

Table 8: Per-class recall (%) for all 9 atlas cells. d2 is the lowest-recall class in every cell. Boldface marks the highest d2 recall.

Cell	d0	d1	d2	d3
Pythia, nuc.	99.3	81.9	59.2	74.4
Pythia, τ	99.2	79.9	56.0	72.1
Pythia, top-k	98.7	71.6	44.2	60.9
OLMo, nuc.	97.8	82.1	60.6	75.3
OLMo, τ	99.3	87.1	65.6	77.1
OLMo, top-k	99.2	80.9	50.4	67.2
GPT-2 XL, nuc.	98.5	87.4	72.8	78.5
GPT-2 XL, τ	99.0	91.1	**73.0**	80.5
GPT-2 XL, top-k	98.7	81.3	54.8	68.4

B.6 Confusion Matrices

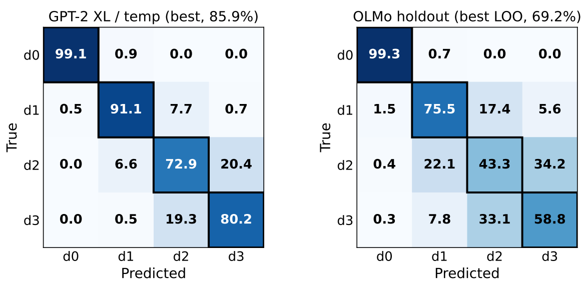

Figure 5: Confusion matrices: best in-distribution (GPT-2 XL temp, 85.9%) and best leave-one-out (LOO) transfer (OLMo, 69.2%). d0 recall >99%; errors spread to d1–d3 under transfer.

B.7 Cross-Strategy Transfer Details

Table 9: Cross-strategy transfer (leave-one-out). Chance = 25% / 33% (d1–d3). d0 recall ≥89% inflates headline; d1–d3 mean: 45.9%.

Holdout	Pythia Acc (%)	OLMo Acc (%)	GPT-2 XL Acc (%)
nucleus	**69.4**	**65.4**	**59.9**
temp (τ=0.9)	62.8	58.2	56.1
top-k (k=50)	51.5	54.9	53.6

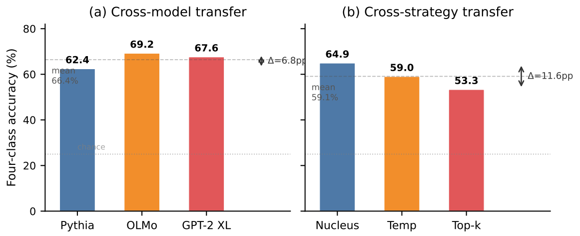

Figure 6: Cross-model vs. cross-strategy transfer accuracy (leave-one-out). Strategy holdout range (Δ=11.6 pp) exceeds model holdout range (Δ=6.8 pp); decoding configuration is the primary transfer barrier.

B.8 Accumulation Pairwise AUC

Table 10 reports pairwise AUC for the six accumulation cells (3 models $\times$ 2 mixing ratios; GPT-2 XL and Pythia use nucleus, OLMo uses temperature 0.9). At 25% mixing, all three cells preserve $\mathrm{EDD}_{0.70}$=3. At 50% mixing, GPT-2 XL nucleus is the sole failure: d2v3 AUC = 0.695, marginally below the 0.70 threshold (0.55 pp).

Table 10: Pairwise AUC under accumulation (GPT-2 XL/Pythia: nucleus; OLMo: temp 0.9; binary RF, 5-fold CV). Bold: $\mathrm{EDD}_{0.70}$ boundary failure.

Model	Mix	d0v1	d1v2	d2v3
GPT-2 XL	25%	1.000	0.817	0.726
	50%	1.000	0.831	**0.695**
Pythia	25%	1.000	0.809	0.749
	50%	1.000	0.793	0.716
OLMo	25%	1.000	0.848	0.759
	50%	1.000	0.817	0.726

B.9 DCOL Iteration Details

Table 11 summarizes all six DCOL iterations. DeBERTa-v3-large exhibited systematic numerical instability across five configurations; RoBERTa-large resolved stability but could not close the accuracy gap with feature-based methods.

Table 11: DCOL iteration summary. All experiments use Pythia-1.4B nucleus data. RF baseline: 78.7%.

Ver.	Backbone	Config	Outcome
v1	DeBERTa	FT, m_lr 10^{-3}	Embed. collapse (31.5%)
v2	DeBERTa	FT, m_lr 10^{-4}	OOM
v3	DeBERTa	FT, m_lr 10^{-4}	Forgetting + OOM
v4	DeBERTa	LoRA, m_lr 10^{-3}	Embed. collapse
v5	DeBERTa	LoRA, m_lr 10^{-4}	NaN gradients
v6	RoBERTa	FT, grad clip 1.0	Stable; 64.5% (−13.7pp)

B.10 Zero-Shot LLM Prompt Template

Both DeepSeek V3.2 and Qwen2.5-7B-Instruct use identical prompting. The system message defines recursive generation depth with per-depth descriptions (depth 0: human-written; depth 1–3: successive LLM generation cycles) and lists expected textual patterns at each depth (vocabulary richness, repetition, coherence). The user message presents the passage (truncated to ≈384 words) and requests a JSON response containing an integer depth prediction (0–3), a confidence score, and a one-sentence reasoning. No in-context examples are provided.

> **System:** "You are an expert at detecting AI-generated text and estimating its generation depth. 'Recursive generation depth' refers to how many times a text has been through an LLM generation cycle: Depth 0: Original human-written text [. . .] Depth 3: Text generated by an LLM given depth-2 text as input. Key patterns by depth: Depth 0 (human): Rich vocabulary [. . .] Depth 3 (third-gen): Noticeable repetition, formulaic structure [. . .]"
>
> **User:** "Analyze the following text and estimate its recursive generation depth (0, 1, 2, or 3). TEXT: {text}. Respond with ONLY a JSON object: {`"depth"`: ⟨int 0–3⟩, `"confidence"`: ⟨float 0–1⟩, `"reasoning"`: ⟨str⟩}"

B.11 Temperature Robustness

The cross-strategy results in §5.3 use a single temperature (τ=0.9). To test whether depth signal generalizes across temperatures, we repeat the classification pipeline at $\tau \in \{0.5, 0.7, 0.9, 1.1\}$ on Pythia-1.4B.

Within-τ robustness. For $\tau \in [0.5, 0.9]$, d1-vs-d2 AUC exceeds 0.80 (peak 0.851 at τ=0.7, comparable to nucleus p=0.95's 0.861) and 4-class accuracy ranges from 75.4% to 77.3% (Table 12); the depth signal is therefore not an artifact of a single temperature. The same top-5 features (`var_surprisal`, `p90_ppl`, `mean_ppl`, `mean_surprisal`, `p75_ppl`) dominate across all τ values with only minor rank-order variation.

Collapse at τ=1.1: memorization as prerequisite. At τ=1.1, 4-class accuracy drops to 57.2% and d1v2 AUC falls to 0.528. Training loss reveals the mechanism: at low temperature (τ=0.5), fine-tuning loss decreases steeply across recursive depths (2.27 $\rightarrow$ 0.42 $\rightarrow$ 0.16), a sign of rapid memorization of the model's own output. At τ=1.1, loss remains nearly flat (2.27 $\rightarrow$ 1.91 $\rightarrow$

1.85); high-entropy output resists memorization, so depth-dependent distributional shifts do not accumulate. This establishes *memorization as a prerequisite* for detectable depth signal: recursive depth is discriminable only when successive fine-tuning iterations progressively reshape the output distribution.

Cross-τ non-transferability. Leave-one-τ-out transfer yields mean accuracy of 46.7% with mean d1v2 AUC of 0.319 (Table 13), worse than cross-strategy transfer (59.1%; §5.3). Holdout τ=0.5 produces inverted d1v2 AUC (0.176); different temperatures therefore produce surprisal distributions with incompatible shapes. This extends the strategy-specificity finding: even within a single decoding family, the feature space is temperature-specific; strategy-conditioned classifiers are therefore necessary.

Ecological validity. On WikiText-103 (546 editorial passages), all three model-specific RF classifiers achieve $\geq$98.5% d0 accuracy despite +621% domain shift in mean perplexity, partly an easy case, as WikiText's higher perplexity places it farther from d1–d3 in feature space. On noisier web-crawl data (5,000 C4 passages), FPR ranges from 4.7% (OLMo) to 22.4% (GPT-2 XL), concentrated in template/SEO content with high n-gram repetition (Table 15); deployment on heterogeneous web data requires model-specific calibration.

Table 12: Within-τ 4-class classification (Pythia-1.4B, 5-fold CV, independent pipeline run). divj: pairwise AUC for depth i vs. depth j. The τ=0.9 accuracy (75.9%) differs from the main atlas (76.8%, Table 2) due to separate generation and LoRA training runs.

τ	4-class acc (%)	d1v2 AUC	d0v1 AUC	d2v3 AUC
0.5	77.3	0.842	0.996	0.764
0.7	75.4	0.851	0.997	0.720
0.9	75.9	0.808	0.991	0.781
1.1	57.2	0.528	0.988	0.599

Table 13: Cross-τ leave-one-out transfer (Pythia-1.4B). Train on three temperatures, test on the held-out one.

Holdout	Acc (%)	d1v2 AUC
τ=0.5	52.2	0.176
τ=0.7	41.0	0.299
τ=0.9	46.4	0.277
τ=1.1	47.1	0.525
Mean	46.7	0.319

B.12 Feature Overlap Analysis

We probe whether end-to-end PLM representations capture the same depth signals as our statistical features via linear probing of [CLS] representations from DeBERTa-v3-base and RoBERTa-base. PLM representations excel at surface lexical features (type–token ratio (TTR): R^2=0.85, repeated bigrams) but fail to capture perplexity distribution statistics; `var_surprisal` and `p90_ppl` yield negative R^2 under linear probing; these features are not linearly accessible in the pretrained representation space. This confirms that the 20pp accuracy gap between feature-based and end-to-end methods (§B.13) stems from complementary, not redundant, representations: PLMs capture lexical patterns while the targeted statistical features capture distributional tail properties that constitute the dominant depth markers.

B.13 End-to-End Learning

DeBERTa-v3-base (He et al., 2021) and RoBERTa-base (Liu et al., 2019) finetuned end-to-end achieve 58.3% (±4.7%) and 59.5% (±3.2%) respectively, lagging the RF baseline (78.7%) by 20.4pp and 19.2pp. The 1.2pp inter-backbone gap rules out architecture bias; Table 14 shows both backbones share the same failure mode (high d0 F1, >0.94, but collapsed d1/d2 performance, F1 $\leq$ 0.42), while RF achieves 0.596 d2 recall, a $>2\times$ advantage at the hardest depth class.

Table 14: Per-class performance for end-to-end PLM classifiers vs. RF baseline (Pythia-1.4B nucleus). PLM columns report F1; RF d2 column reports recall (the class with the largest PLM–RF gap). Both PLM backbones share the same failure pattern: collapsed d1/d2 scores despite adequate d0/d3.

Method	Acc (%)	d0	d1	d2	d3
DeBERTa-v3	58.3 ± 4.7	.947	.378	.247	.656
RoBERTa-base	59.5 ± 3.2	.958	.416	.261	.647
RF (19 feat.)	78.7	—	—	.596†	—

†Recall; shown for the depth class with the largest PLM–RF gap.

Feature space divergence. SHAP TreeExplainer analysis confirms that perplexity and surprisal features account for 88.8% of RF's total contribution. Three independent importance measures (SHAP, Gini, and permutation) show strong concordance (Kendall's W=0.90, p<0.005; pairwise Spearman ρ: SHAP–Gini 0.93, SHAP–permutation 0.84; §B.2), with `p90_ppl` appearing in the SHAP top-2 in 9/9 cells and `var_surprisal` in 8/9.

Linear probing of PLM [CLS] representations yields negative R^2 for both `var_surprisal` and `p90_ppl` (§B.12); pretrained representations lack the linear structure needed to recover perplexity tail statistics, a systematic representational gap rather than a backbone-specific artifact.

DCOL (Pitawela et al., 2025) reached 64.5% (§B.9); all end-to-end approaches (58–65%) underperform feature-based methods; depth estimation therefore benefits from targeted statistical features that capture distributional tail properties absent from PLM representations. This comparison is bounded by backbone scale (≤125M parameters); whether larger PLMs (e.g., 7B+) can close the gap via richer internal representations remains an open question.

B.14 Domain Bias Analysis

To assess whether the classifier exploits depth-related features or memorizes stylistic cues from individual seed articles, we conduct three sub-experiments.

Methodological note. An initial implementation assigned sample IDs with a global counter across depths, inadvertently correlating seed-group identity with depth (LOGO accuracy 64.0%, seed-group prediction 20.8%). We discovered the issue during internal review, corrected to per-depth seed assignment ensuring each group contains equal samples per depth, and re-ran all sub-experiments. The results below use the corrected pipeline; the initial (buggy) numbers are reported for transparency.

Sub-A: Standard CV. Five-fold CV accuracy on 3-class (d1/d2/d3) classification ranges from 59% to 82% across the 9 atlas cells (up to 2.5× above the 33.3% chance level); depth signal is therefore the dominant classification factor.

Sub-B: LOGO evaluation. We assign each passage to one of 10 seed-article groups (balanced: each group contains equal samples per depth) and perform leave-one-group-out CV. LOGO mean accuracy is 72.1% (range 59.5–81.6%), matching standard CV (72.0%) within noise (before fix: 64.0%, reflecting the depth–group confound). Per-cell fold standard deviations range from 0.4pp to 1.4pp; cross-group generalization is therefore as reliable as within-group classification.

Sub-C: Negative control. We train two classifiers on identical features: one predicting depth, the other predicting seed-group identity (10-class, chance = 10%). Depth prediction accuracy averages 72.0%; seed-group prediction averages 10.2%, indistinguishable from chance (before fix: 20.8%, inflated by the depth–group confound). The depth-to-seed-group accuracy ratio is 7.1×; the feature representation therefore encodes depth but not seed-article identity.

Taken together, these results show that seed-article grouping has no measurable effect on depth estimation after correcting the group assignment: LOGO accuracy equals standard CV, and seed-group prediction is at chance.

B.15 Wild-Data Calibration

Table 15 reports d0 accuracy and false positive rate (FPR) for all three model-specific nucleus RF classifiers on two out-of-distribution human-text corpora: WikiText-103 (546 editorial passages) and C4 (5,000 web-crawl passages, crawled 2019–2020; pre-LLM deployment).

Table 15: Wild-data d0 accuracy and FPR for nucleus RF classifiers. All passages are human-written (true d0). FPR = fraction misclassified as d1+. 95% Wilson CIs computed for C4 (n=5,000).

Model	Source	n	d0 (%)	FPR (%)	95% Wilson CI
Pythia-1.4B	WikiText	546	100.0	0.0	—
	C4	5,000	79.8	20.2	[19.1, 21.3]
GPT-2 XL	WikiText	546	99.3	0.7	—
	C4	5,000	77.6	22.4	[21.3, 23.6]
OLMo-1B	WikiText	546	98.5	1.5	—
	C4	5,000	95.3	4.7	[4.2, 5.3]

On WikiText-103, all three classifiers achieve ≥98.5% d0 accuracy; editorial text is therefore reliably distinguished from recursive generations across model families. On C4, FPR varies sharply: OLMo (4.7%) vs. Pythia (20.2%) and GPT-2 XL (22.4%). C4 misclassifications concentrate in template and SEO content with elevated repetition features (`rep_3gram` +37%, `rep_4gram` +75% vs. correctly classified d0 passages), rather than in random noise. The model gap likely reflects training data composition: OLMo (trained on Dolma, which includes diverse web data) produces d0 features more tolerant of web-crawl variation, while Pythia and GPT-2 XL (trained primarily on Wikipedia/WebText) map template-heavy content into the d1+ region. Among C4 false positives for GPT-2 XL, predictions distribute as d1 (9.0%), d2 (3.7%), d3 (9.6%); for OLMo, d1 (2.9%), d2 (0.5%), d3 (1.3%). The non-uniform depth distribution suggests the classifier maps specific repetition patterns to particular depth signatures rather than producing random misclassifications.

Threshold calibration. A pooled RF classifier (trained on all three families' nucleus data) achieves Expected Calibration Error (ECE) = 0.052 on in-distribution data; Platt scaling and isotonic regression yield no improvement; probability estimates are already well calibrated. The high FPR on web-crawl data is therefore a decision-threshold problem, not a probability calibration defect. Applying a confidence threshold $P(\mathrm{d0}) \geq 0.15$ to the pooled classifier reduces its C4 FPR from 33.4% (not shown in Table 15) to 2.8%, at a cost of only -0.3 pp balanced accuracy (78.0%→77.7%). The $P(\mathrm{d0})$ distribution is bimodal: 67% of samples exceed 0.95 (high-confidence d0), while 33% fall in the 0.15–0.30 uncertainty zone; the threshold separates these modes cleanly. Wild-data FPR is thus an operational threshold choice rather than a fundamental limitation, though the optimal threshold is domain-dependent.

Additional web corpora. Evaluating the pooled classifier on FineWeb (10,000 passages) and RedPajama (10,000 passages) yields raw FPR of 17.2% and 15.5%, respectively, both below C4 (34.0%), likely reflecting less template-heavy content in these corpora. Posterior-threshold calibration ($P(\mathrm{d0}) \geq 0.10$) reduces FPR to 0.1% (FineWeb) and 7.0% (RedPajama); across all three web-crawl corpora, calibrated FPR remains $\leq$7%.

B.16 Downstream Curation PoC (Inconclusive)

To assess whether depth estimation translates to actionable data curation, we conduct a proof-of-concept downstream evaluation. We fine-tune Pythia-410M with LoRA (r=16, α=32) for 3 epochs on 8 data conditions derived from the Pythia nucleus depth-estimation corpus, repeating each condition over 5 random seeds. Conditions vary the selection mechanism (RF-predicted depth, oracle depth, random subsampling) and stringency (strict ≈25%, moderate ≈50%, soft threshold). Table 16 reports per-condition perplexity; Table 17 reports pairwise contrasts.

For the binary task of identifying low-depth passages (d0 vs. d1–d3, evaluated on the same Pythia nucleus corpus with 5-fold CV), RF-based depth filtering achieves F1 = 0.997, substantially outperforming a simple perplexity-threshold baseline (F1 = 0.845); the multi-feature representation provides discriminative value beyond raw perplexity. However, the primary comparison (RF-filtered vs. unfiltered, $p > 0.4$) is non-significant; the significant RF-vs-random contrasts ($p < 0.02$) involve multiple pairwise comparisons that were not corrected for multiplicity.

Table 16: Curation PoC: per-condition perplexity (Pythia-410M LoRA, 5 seeds). Δ is signed so that positive values indicate lower (better) perplexity than unfiltered. d_z: Cohen's paired effect size. $^*: p < 0.05$ (paired t-test vs. unfiltered).

Condition	Mean PPL	Std	n	Δ vs Unfilt.	p	d_z
unfiltered	15.776	0.066	10,000	—	—	—
rf_soft	15.733	0.038	4,422	+0.043	0.407	0.41
rf_moderate	15.756	0.056	4,940	+0.021	0.492	0.34
rf_strict	15.810	0.061	2,496	−0.034	0.607	−0.25
oracle_moderate	15.809	0.103	5,000	−0.033	0.485	−0.34
oracle_strict	15.879	0.128	2,500	−0.103	0.216	−0.66
random_soft	16.062	0.079	4,422	−0.286	<0.001*	−2.77
random_moderate	15.998	0.090	4,940	−0.222	0.028*	−1.50
random_strict	16.239	0.165	2,496	−0.463	0.013*	−1.89

Table 17: Pairwise comparisons (paired t-test, 5 seeds). $^*: p < 0.05$; $^{**}: p < 0.01$.

Comparison	Δ PPL	t	p
rf_strict vs random_strict	0.429	−7.05	0.002**
rf_moderate vs random_moderate	0.242	−3.77	0.020*
rf_soft vs random_soft	0.328	−17.98	<0.001**
rf_soft vs unfiltered	0.043	0.93	0.407

B.17 Reference Model Ablation

Table 18 ablates the reference model used for feature extraction on Pythia-1.4B nucleus text (d0–d3, 5-fold CV). Self-reference (Pythia-1.4B) yields 78.6%; switching to a cross-architecture reference degrades accuracy by 4–11 pp, confirming that reference-generator alignment amplifies depth signal.

Table 18: Reference model ablation. Classification accuracy (5-fold CV) on Pythia-1.4B generated text (nucleus, d0–d3) using different reference models for feature extraction.

Reference Model	Architecture	Accuracy (%)
Pythia-1.4B (self)	GPT-NeoX	78.6 ± 0.6
OLMo-1B (cross)	OLMo	74.5 ± 0.7
GPT-2 XL (cross)	GPT-2	67.3 ± 0.7

B.18 Cross-Scale Full Results

This section provides the full cross-scale analysis summarized in §5.4. We extend the benchmark to two 7–8B models (Qwen2.5-7B-Instruct (Qwen Team, 2024), Llama-3.1-8B-Instruct (Grattafiori et al., 2024)) and one 14B model (Qwen2.5-14B). The 7–8B models use 4-bit NF4 QLoRA fine-tuning with nucleus sampling (p=0.95, 500 samples per depth, depths $d \in \{0, 1, \ldots, 5\}$). The 14B model uses clean FP16 LoRA fine-tuning across all three decoding strategies (nucleus, temperature τ=0.9, top-k k=50; 5,000 samples per depth, depths $d \in \{0, 1, 2, 3\}$). Feature extraction for 1–8B models uses Pythia-1.4B as reference; Qwen2.5-14B is evaluated under both self-reference and cross-reference (Pythia-1.4B) to disentangle reference-model artifacts from scale effects (see paragraph 5 below).

Scale comparison. Table 19 contrasts the 7–14B models with the three 1–1.5B models on matched evaluation (depths 0–3). The 14B model is tested across all three decoding strategies; the 7–8B models use nucleus sampling only.

Table 19: Cross-scale comparison: 4-class RF accuracy (%, depths 0–3), $\mathrm{EDD}_{0.70}$ (computed from full depth range: 0–5 for 1–1.5B nucleus and 7–8B, 0–3 for 1–1.5B temp/top-k and 14B), and d0-vs-d1 AUC. CI: 95% bootstrap. Chance = 25%. †Self-reference model (see text).

Model	Strategy	4-class acc	$\mathrm{EDD}_{0.70}$	d0v1
GPT-2 XL (1.5B)	nucleus	84.3	4	0.999
OLMo-1B	nucleus	78.9	4	0.997
Pythia-1.4B	nucleus	78.7	4	0.998
Llama-3.1-8B	nucleus	70.2 [68.3, 72.4]	4	0.949
Qwen2.5-7B	nucleus	60.1 [58.0, 62.1]	5	0.969
Qwen2.5-14B†	nucleus	54.1 [53.4, 54.7]	0	0.643
Qwen2.5-14B†	temp09	50.5 [49.8, 51.2]	0	0.632
Qwen2.5-14B†	top-k50	47.9 [47.3, 48.6]	0	0.561
14B mean		*50.8*	—	—

(1) Depth signal persists at 7–8B scale. Both 7–8B models substantially exceed the 25% chance baseline: Llama-3.1-8B achieves 70.2% (95% CI $[68.3, 72.4]$) and Qwen2.5-7B achieves 60.1% ($[58.0, 62.1]$); recursive depth therefore remains estimable from perplexity-based features even when the generator is 5–7× larger than the reference model. Depth-0 separability is preserved in both models: d0v1 AUC is 0.949 (Llama-3.1) and 0.969 (Qwen2.5).

(2) EDD at 7–8B scale. At threshold T=0.70, Qwen2.5-7B maintains all consecutive-pair AUCs above 0.70 through depth 5 (Table 20), yielding $\mathrm{EDD}_{0.70} = 5$ (the maximum observable from the tested depth range), exceeding the $\mathrm{EDD}_{0.70}$=4 observed at 1–1.5B. Llama-3.1-8B matches the smaller models at $\mathrm{EDD}_{0.70}$=4: its monotonic decay causes d4v5 (0.618) to drop below T, despite stronger signal at shallower depths. Both larger generators produce weaker per-step signal than the 1–1.5B models (mean consecutive AUC: 0.786 for Llama-3.1, 0.782 for Qwen2.5, vs. 0.852 at 1–1.5B).

Table 20: Consecutive binary AUC. Llama-3.1-8B exhibits monotonic decay; Qwen2.5-7B shows a U-shaped pattern; Qwen2.5-14B† (depths 0–3 only) has d0v1 below T=0.70 in all strategies, yielding $\mathrm{EDD}_{0.70}$=0 despite some later pairs exceeding the threshold. †Self-reference model.

Model	Strategy	d0v1	d1v2	d2v3	d3v4	d4v5
Llama-3.1-8B	nucleus	0.949	0.877	0.771	0.716	0.618
Qwen2.5-7B	nucleus	0.969	0.729	0.709	0.732	0.772
Qwen2.5-14B†	nucleus	0.643	0.713	**0.728**	–	
Qwen2.5-14B†	temp09	0.632	**0.726**	0.618	–	
Qwen2.5-14B†	top-k50	0.561	0.687	0.679	–	

(3) Divergent AUC trajectories and instruction-tuning effect. Among instruction-tuned models, Llama-3.1-8B exhibits monotonic AUC decay consistent with the 1–1.5B pattern, while Qwen2.5-7B shows a U-shaped recovery beyond d2v3 (Table 20). A base-model control (Llama-3.1-8B without instruction tuning) clarifies the causal direction: the base model achieves 59.9% four-class accuracy (95% CI [57.8, 62.15], $\mathrm{EDD}_{0.70}$=2) with non-monotonic AUC decay (Table 21), while the instruction-tuned variant reaches 70.2% with strictly monotonic decay. The base-model control experiment suggests instruction-tuning regularizes rather than confounds the depth signal. Separately, the FP16 self-reference experiment at 7B (Appendix B.19) produces strictly monotonic AUC decay (0.697, 0.673, 0.659), indicating that the U-shaped pattern in QLoRA Qwen2.5-7B likely originates from the cross-reference setup (Pythia-1.4B as observer) rather than from an intrinsic property of the depth signal.

Table 21: Pairwise AUC: Llama-3.1-8B Base vs. Instruct.

Variant	d_{0v1}	d_{1v2}	d_{2v3}	d_{3v4}	d_{4v5}
Base	0.918	0.794	0.682	0.730	0.595
Instruct	0.949	0.877	0.771	0.716	0.618

The base model exhibits non-monotonic decay with a U-shaped recovery at d_{3v4}, while the instruction-tuned variant shows strictly monotonic decay. Both variants exceed chance (AUC > 0.5) at all boundaries; depth signal exists regardless of instruction-tuning, though instruction-tuning strengthens and regularizes it.

(4) Signal attenuation at 14B. Qwen2.5-14B (clean FP16, self-reference) is tested across all three decoding strategies (Table 19). Mean four-class accuracy is 50.8% (2.0× chance), with nucleus 54.1%, temperature 50.5%, and top-k 47.9%. $\mathrm{EDD}_{0.70}$=0 across all three strategies because the d0/d1 boundary falls below the 0.70 AUC threshold in every case (0.56–0.64), breaking the contiguous prefix at the first boundary. The d0/d1 boundary, near-perfect at 1–1.5B (AUC >0.99), attenuates to 0.56–0.64 at 14B.

Each strategy produces a structurally distinct adjacent-pair AUC pattern (Table 20): nucleus is monotonically increasing (d0v1=0.643 < d1v2=0.713 < d2v3=0.728); temperature exhibits d2v3 collapse (0.618, the weakest boundary) because temperature scaling uniformly amplifies logit differences, accelerating convergence to the attractor distribution at deeper depths; top-k exhibits d0v1 collapse (0.561, the weakest boundary) because vocabulary truncation constrains d0 and d1 distributions to the same top-k tokens, compressing their statistical distance. This three-way divergence confirms at 14B that different decoding strategies induce structurally distinct degradation trajectories (§5.3).

Feature importance shifts under top-k: lexical diversity metrics (TTR, hapax ratio) enter the top-5, displacing perplexity tail statistics that dominate under nucleus and temperature. This result carries an important caveat: Qwen2.5-14B uses self-reference, whereas all other experiments use Pythia-1.4B. Self-reference conflates the generator's distributional shift with the reference model's sensitivity; the observed attenuation may partially reflect the reference model change rather than a pure scale effect.

(5) Cross-reference control at 14B. To disentangle the scale effect from the reference-model artifact, we re-extract all 19 features using Pythia-1.4B (the same reference model as the 1–1.5B benchmark) on the Qwen2.5-14B generated text. Table 22 compares self-reference and cross-reference results per strategy.

Cross-reference accuracy drops by 6.8 pp on average (range 4.2–9.1 pp), but $\mathrm{EDD}_{0.70}$ *rises* from 0 to 2 for nucleus and temperature: the consecutive d0v1 (0.70–0.74) and d1v2 (0.72–0.73) AUCs now exceed the 0.70 threshold, while d2v3 remains below (0.54–0.61). Under self-reference, the first boundary d0v1 falls below T=0.70 in all strategies (0.56–0.64), breaking the contiguous prefix and yielding $\mathrm{EDD}_{0.70}$=0 even though some later pairs (e.g., nucleus d1v2 = 0.713, d2v3 = 0.728) exceed the threshold; the Pythia-1.4B reference model provides sharper distributional contrast at the cost of overall accuracy. The ∼7 pp accuracy gap between self- and cross-reference is therefore a reference-model artifact, not a depth-signal artifact: depth signal persists at 14B albeit attenuated relative to 1–1.5B.

Table 22: 14B cross-reference control: Pythia-1.4B reference vs. self-reference on Qwen2.5-14B text. Δ: cross − self. $\mathrm{EDD}_{0.70}$ computed from consecutive pairwise AUC.

Strategy	Self-ref (%)	Cross-ref (%)	Δ	$\mathrm{EDD}_{0.70}$	d0v1 AUC
nucleus	54.1	45.0	−9.1	2	0.702
temp (τ=0.9)	50.5	46.3	−4.2	2	0.739
top-k (k=50)	47.9	40.9	−7.1	0	0.699
mean	*50.8*	*44.1*	*−6.8*	—	—

Caveats. For Qwen2.5-7B, the CI lower bound (58.0%) does not exclude sub-60% accuracy; Llama-3.1-8B's lower bound (68.3%) is more comfortably above chance but still below the 75–83% range at 1–1.5B. The divergent AUC trajectories across two models caution against strong generalizations from only two 7–8B data points: both the monotonic (Llama-3.1) and non-monotonic (Qwen2.5) patterns are single observations, and further architectures are needed to establish which is the dominant regime at this scale.

Confounds. Four confounds limit the 7–8B results: (1) 4-bit NF4 QLoRA quantization (absent at 1–1.5B) may distort the perplexity distribution independently of depth; (2) smaller sample size (500 vs. 5,000 per depth) widens confidence intervals; (3) evaluation is limited to nucleus sampling at 7–8B (the 14B model addresses this with all three strategies); (4) vocabulary mismatch between larger generators and the 1.4B reference model. For the 14B model specifically: the cross-reference control (Table 22) quantifies the reference-model artifact at ∼7 pp, but the remaining accuracy gap (44% vs. 78–85% at 1–1.5B) still conflates scale with the other confounds listed above. The base-model control partially addresses a former confound (instruction-tuning): since the base variant shows weaker signal with non-monotonic decay while the instruction-tuned variant shows stronger monotonic signal, instruction-tuning regularizes rather than confounds depth estimation. The remaining confounds are not mutually independent (quantization and vocabulary mismatch both affect the perplexity distribution); it is therefore infeasible to attribute the observed accuracy gap (60–70% vs. 78–85% at 1–1.5B) to any single factor.

B.19 Qwen2.5-7B Self-Reference Scale Probe

The preceding 7–8B experiments (§B.18) use 4-bit NF4 QLoRA with Pythia-1.4B cross-reference, leaving two confounds unresolved: (1) quantization-induced perplexity distortion and (2) vocabulary mismatch between the 7B generator and the 1.4B reference model. To isolate the scale effect from these artifacts, we run the full recursive generation pipeline on Qwen2.5-7B (Qwen Team, 2024) (base model, not instruction-tuned) using FP16 LoRA fine-tuning and self-reference feature extraction (the same model serves as both generator and reference).

Setup. At each depth $k \in \{0, 1, 2, 3\}$, a fresh LoRA adapter (r=16, α=32, dropout 0.05, target modules `q_proj`, `k_proj`, `v_proj`) is initialized on the base model and trained on $\mathcal{D}_{k-1}$ for 3 epochs (lr $2{\times}10^{-4}$, effective batch size 16 via 4×4 gradient accumulation, bfloat16 mixed precision), following the same protocol as the 1–1.5B benchmark (§3.2). Generation uses nucleus sampling (p=0.95), producing 5,000 passages per depth (20,000 total, matching the per-cell sample size of the primary benchmark). Feature extraction computes the standard 19-dimensional feature vector $\mathbf{f}(x)$ using the model's own token-level log-probabilities as the reference distribution, eliminating the cross-model reference artifact quantified in Table 22. Classification uses the same 5-fold stratified Random Forest (200 trees, seed 42) as all other cells.

Comparison to prior 7B experiments. This experiment differs from the QLoRA Qwen2.5-7B-Instruct entry in Table 19 in three respects: (i) FP16 LoRA removes quantization distortion, (ii) self-reference removes vocabulary mismatch, and (iii) 5,000 samples per depth (vs. 500) narrows confidence intervals. If depth signal persists under these cleaner conditions, the attenuation observed in the QLoRA experiments reflects genuine scale effects rather than methodological artifacts.

Results. The self-reference probe achieves 52.1% four-class accuracy (CI [51.4, 52.7], 2.1× chance, $\mathrm{EDD}_{0.70}$=0; $\mathrm{EDD}_{0.65}$=3). Pairwise AUC: d0-vs-d1 = 0.697, d1-vs-d2 = 0.673, d2-vs-d3 = 0.659; top-5 features: `p90_ppl`, `skew_ppl`, `p90_surprisal`, `kurt_ppl`, `var_surprisal`. Contrary to the hypothesis that removing quantization and vocabulary-mismatch confounds would improve accuracy, self-reference at 7B is 8 pp below the QLoRA cross-reference baseline (60.1%), and closely matches the 14B self-reference result (50.8%, $\mathrm{EDD}_{0.70}$=0). This reveals a self-reference ceiling: larger models assign similar probabilities to d0 and d1 text, collapsing d0v1 discriminability (AUC = 0.697 vs. 0.969 under cross-reference). Cross-model reference strategies, where a smaller observer model provides complementary distributional perspective, may be necessary for effective depth estimation above ~7B.

B.20 Cross-Boundary Generalization

The preceding experiments assume a single model family generates all recursive depths. In practice, recursive contamination may traverse model boundaries: text generated by model A is used to train model B, whose output trains model C. We probe whether the depth signal persists under two relaxations of the single-model assumption.

Cross-model recursive chains. We construct a heterogeneous chain: human text (d0) → Pythia-1.4B generates d1 → OLMo-1B finetuned on Pythia-d1 generates d2 → GPT-2 XL finetuned on OLMo-d2 generates d3. Each recursive step uses a *different* model family, eliminating the possibility that depth signal arises from a single model's idiosyncratic behavior. A 4-class Random Forest achieves 84.75% accuracy (95% CI $[81.0, 88.25]$), with per-depth F1 scores of 0.98 (d0), 0.88 (d1), 0.76 (d2), and 0.76 (d3). Pairwise AUC remains high: d0-vs-d1 = 0.999 and d2-vs-d3 = 0.841. These results suggest that distributional artifacts of recursive generation persist across model boundaries. This constitutes preliminary evidence for a process-level depth signal rather than individual generator artifacts.

Confound discussion. The heterogeneous chain confounds architecture identity with depth (d1=Pythia, d2=OLMo, d3=GPT-2 XL). However, if the classifier exploited architecture fingerprints rather than depth signal, near-perfect accuracy would be expected since architecture perfectly determines depth label. The observed 84.75% with monotonically decreasing per-depth F1 (d0=0.98, d1=0.88, d2=0.76, d3=0.76) is more consistent with a depth-signal interpretation, as architecture fingerprinting would not produce this graded pattern. We address this confound directly with a reverse-chain control below.

Reverse-chain control. To empirically test the architecture confound hypothesis, we constructed a reverse chain (GPT-2 XL $\rightarrow$ OLMo $\rightarrow$ Pythia) reversing the model-depth assignment. The reverse chain achieves 85.75% (95% CI [82.0, 89.25]), statistically indistinguishable from the forward chain (84.75%, [81.0, 88.25]). If the classifier exploited architecture fingerprints rather than depth signal, reversing the chain should invert depth predictions; the equivalent accuracy under both orderings provides empirical evidence against the architecture confound.

Cross-domain transfer. All prior experiments use WikiText-103 as the seed corpus. To test domain dependence, we repeat the pipeline with arXiv abstracts as the seed text (Pythia-1.4B, nucleus p=0.95, 500 samples per depth, d0–d3). Five-fold cross-validation yields 74.1% $\pm$ 2.0% accuracy, compared to $\sim$78% on the WikiText seed. Per-depth F1 shows the familiar pattern: d0 = 1.00, d1 = 0.78, d2 = 0.52, d3 = 0.66, with d2 remaining the most challenging class. The 4pp accuracy gap relative to WikiText likely reflects the narrower stylistic range of scientific abstracts, which may reduce the distributional contrast between adjacent depths. These results provide preliminary evidence that the perplexity-based features capture properties of the recursive generation process rather than domain-specific artifacts of the seed corpus.

B.21 Cross-Seed Replication

Table 23 reports within-seed accuracy for all three model families (nucleus, depths 0–3) under two independent LoRA training seeds.

Seed=43 yields higher accuracy in all three models (+0.14 to +3.17 pp, same direction). For Pythia (+0.14 pp), CIs overlap, consistent with noise; for OLMo (+3.17 pp) and GPT-2 XL (+2.04 pp), CIs do not overlap, so LoRA initialization has a non-trivial effect on absolute accuracy ($\sim$1–3 pp). Top-3 features are identical across all seeds and models (`p90_ppl`, `var_surprisal`, `p75_ppl`); depth ordering is preserved in every configuration.

Table 23: Cross-seed replication (nucleus p=0.95, d0–d3). seed=42 is the primary benchmark (matching Table 2); seed=43 is an independent replication.

Model	Seed	Acc (%)	95% CI	Δ (pp)
Pythia-1.4B	42	78.70	[78.10, 79.24]	—
Pythia-1.4B	43	78.84	[78.13, 79.52]	+0.14
OLMo-1B	42	78.94	[78.38, 79.48]	—
OLMo-1B	43	82.11	[81.77, 82.54]	+3.17
GPT-2 XL	42	84.28	[83.78, 84.80]	—
GPT-2 XL	43	86.32	[86.17, 86.49]	+2.04

Outlier handling. Feature extraction occasionally produces extreme perplexity values (outlier passages); we apply per-feature clipping at $\pm 10^{10}$ before classification, affecting $<0.1\%$ of samples. The d3 perplexity distribution varies substantially across seeds: under seed=42, 0.56% of d3 passages exceed 10^6 in raw perplexity, while under seed=43 this fraction rises to 12.8%. Despite this variation, the classifier remains effective after clipping, confirming that the depth signal resides in the p90-and-below distributional shape rather than extreme outliers.

B.22 Depth-4 Extension

We extend all three model families (GPT-2 XL, Pythia-1.4B, OLMo-1B; nucleus p=0.95) to depth 4 and train 5-class RF classifiers (d0–d4, 5,000 samples per depth). Five-class accuracy reaches 80.5% (OLMo), 74.0% (GPT-2 XL), and 71.3% (Pythia), all substantially above the 20% chance baseline. For Pythia and GPT-2 XL, d3 recall drops from $\sim$80% (4-class) to $\sim$49% (5-class) due to d3/d4 confusion; d4 recall is higher (Pythia 68.1%, GPT-2 XL 63.9%). $\mathrm{EDD}_{0.70} \geq 4$ was established by dedicated binary classifiers in the atlas (§4.3; d3v4 AUC: Pythia 0.784, OLMo 0.765, GPT-2 XL 0.750); the 5-class results provide complementary evidence that depth 4 remains discriminable in the multi-class setting.

Feature monotonicity at d4. For Pythia and GPT-2 XL, the dominant depth markers (`p90_ppl`, `var_surprisal`) maintain monotonic decrease through d4 (Pythia `p90_ppl`: 483.7 $\rightarrow$ 12.5 from d0 to d4; `var_surprisal`: 6.50 $\rightarrow$ 1.29), consistent with continued tail erosion (§3.5). Mean perplexity breaks monotonicity at d4 for GPT-2 XL (d3: 1,076K $\rightarrow$ d4: 928K) while tail perplexity continues to decline. OLMo exhibits a `var_surprisal` rebound at d4 (2.49 → 2.83, non-monotonic), creating a feature signature absent in GPT-2 XL and Pythia. The 5-class RF exploits non-adjacent depth contrasts (particularly d0-vs-d4 and d2-vs-d4) to achieve 80.5% accuracy, the highest among all three families, while the dedicated binary d3-vs-d4 classifier (AUC 0.765) is restricted to adjacent-depth features. This discrepancy illustrates that multi-class classifiers can exploit global distributional structure unavailable to pairwise comparisons. The divergence between aggregate and tail statistics suggests that aggregate distributional statistics saturate at deep recursion while tail statistics continue to shift. This saturation provides a physical mechanism for the EDD ceiling: once per-step feature change falls below classifier resolution, adjacent depths become indistinguishable regardless of continued distributional drift.

B.23 Generation Length Ablation

Table 24 reports classification accuracy as a function of passage length (Pythia-1.4B nucleus, depths 0–3).

Table 24: Generation length ablation (Pythia-1.4B nucleus, 5-fold CV). Accuracy increases monotonically with passage length; the 9.47 pp spread confirms that longer passages provide stronger statistical signal.

Passage length (tokens)	Acc (%)
128	68.35
256	73.80
512	77.82

Depth signal is detectable from 128 tokens (2.7× chance) and saturates near 500 tokens. The monotonic increase reflects the statistical nature of the features: longer passages yield more stable estimates of perplexity percentiles and surprisal variance, reducing within-class overlap at adjacent depth boundaries.

B.24 Mixed-Depth Corpus

Per-passage classification is largely independent of corpus composition: evaluating on a corpus containing mixed proportions of d0–d3 passages (rather than balanced classes) yields macro-recall $\Delta = -1.62$ pp relative to the balanced benchmark (within the 2 pp tolerance). Individual class deltas range from -0.45 pp to -2.43 pp (d1 shows the largest shift); the RF classifier operates on per-sample features without sensitivity to the class distribution of the evaluation corpus, as Random Forest produces identical per-sample predictions regardless of batch context.

B.25 Learned Estimator Comparison

Table 25 evaluates three learned classifiers (XGBoost, ordinal MLP with CORAL loss (Cao et al., 2020), prototypical networks) under the same leave-one-model-out protocol as Table 4. The ordinal MLP follows the nested binary reduction framework (McCullagh, 1980; Frank and Hall, 2001) with the loss function of Rennie and Srebro (2005). All three methods operate on the same 19-dimensional feature vector; the comparison isolates the effect of the classification head.

Table 25: Learned estimator cross-model transfer (leave-one-model-out, nucleus, 4-class). RF baseline from Table 4. Chance = 25%.

Method	Pythia	OLMo	GPT-2 XL	Mean
RF (baseline)	—	—	—	66.4
XGBoost	63.7	69.8	68.1	67.2
Ordinal MLP	62.5	69.7	67.9	66.7
Prototypical	61.6	70.0	68.4	66.6

All three learned methods match the RF baseline within 0.8 pp on average; the Pythia holdout is consistently the hardest (61.6–63.7%), while OLMo and GPT-2 XL holdouts yield 67.9–70.0%. The narrow spread ($\leq$0.8 pp across four methods) reinforces the conclusion that the 19-feature representation, not the classifier, determines the accuracy ceiling in the cross-model transfer setting.

B.26 Instruction-Tuned Model at 1.5B Scale

To test whether instruction tuning eliminates depth signal at matched scale (1.5B, avoiding the quantization confound of §5.4), we run the full recursive pipeline on Qwen2.5-1.5B-Instruct with self-reference feature extraction (i.e., the same model serves as both generator and reference).

The RF classifier achieves 47.4% four-class accuracy (95% CI $[46.7, 48.1]$, 1.9× chance); instruction tuning therefore reduces but does not eliminate depth signal. The pairwise AUC profile differs from base models: d0v1 = 0.653, d1v2 = 0.620, d2v3 = 0.664, with a gap-dependent pattern (non-adjacent pairs yield higher AUC, e.g., d0v3 = 0.902) but no pair reaching the 0.75 threshold.

The feature importance ranking shifts: the perplexity tail statistics (`p90_ppl`, `var_surprisal`) that dominate in base models drop in relative importance, while surprisal variance and mean-level features gain rank. This shift suggests instruction tuning compresses the perplexity distribution (reducing tail discriminability) while preserving lower-order statistical patterns.

Two caveats apply: (1) self-reference evaluation may inflate or deflate accuracy relative to cross-reference evaluation (cf. §B.17); (2) a single instruction-tuned model does not establish the generality of the feature shift across architectures or instruction-tuning recipes.

B.27 Downstream Fine-Tuning Probe (Inconclusive)

We fine-tune Pythia-410M with LoRA (r=16, α=32) on depth-stratified subsets (500 samples each) and evaluate held-out perplexity (200 samples, 50 per depth).

Table 26: Downstream fine-tuning probe: held-out perplexity by training subset (Pythia-410M LoRA, seed=42). Δ: signed difference vs. baseline (negative = better).

Training subset	PPL	Δ vs. baseline
Baseline (no fine-tuning)	5.67	—
d0 only	5.65	−0.02
d1 only	5.55	−0.12
d2 only	5.56	−0.11
Mixed (d0–d3)	5.55	−0.12

The hypothesis that d0-only data yields the lowest perplexity is not supported: d1-only and mixed subsets tie for best (PPL 5.55), both marginally below d0-only (5.65). The PPL spread is small (0.12, 2.1% relative) and the sample size (500 per condition, single seed) is insufficient for statistical significance. Training loss decreases monotonically with depth (d0: 2.558, d1: 1.920, d2: 1.286), consistent with recursive generation reducing information content and making text easier to model. This pilot establishes that depth-filtered data is not harmful for downstream fine-tuning; whether depth-aware curation provides a meaningful advantage requires larger-scale evaluation.

B.28 Feature Group Ablation

Table 27 reports in-distribution group ablation accuracy (5-fold CV, mean across all 9 model–strategy cells). Table 28 reports the same ablation under cross-model transfer (leave-one-model-out, nucleus).

Table 27: Feature group ablation (in-distribution, 9-cell mean, 5-fold CV, 4-class). Chance = 25%.

Feature group	#Feat.	Acc (%)	Δ vs. Full
Full (all 19)	19	78.4	—
Perplexity only	9	76.6$\pm$5.6	−1.8
Surprisal only	3	73.2$\pm$5.7	−5.2
Lexical Diversity only	2	46.1$\pm$4.5	−32.3
N-gram Entropy only	2	39.5$\pm$5.7	−38.9
Repetition only	3	43.7$\pm$5.2	−34.7

Table 28: Feature group ablation (cross-model LOO, nucleus, 4-class). Groups merge the four Table 1 categories into three: bigram/trigram entropy move from Lexical Diversity to Entropy/Surprisal; TTR/hapax move to Repetition/Diversity. Chance = 25%.

Feature group	#Feat.	Acc (%)	Δ vs. Full
Full (all 19)	19	66.4	—
Perplexity only	9	66.0	−0.4
Entropy/Surprisal only	5	63.2	−3.2
Repetition/Diversity only	5	53.0	−13.4
`p90_ppl` only	1	58.3	−8.1

Under both settings perplexity features nearly match the full set, while repetition/diversity features fall far below; perplexity degradation statistics carry the dominant depth signal. The top-2 individual features by Gini importance are `p90_ppl` (22.4%) and `var_surprisal` (18.8%), accounting for 41% of total importance.

B.29 RF Seed Robustness

Table 29 reports cross-model transfer accuracy across three RF random seeds.

Table 29: RF seed robustness (cross-model LOO, nucleus, 4-class).

Seed	Acc (%)
42	66.4
123	66.5
456	66.5
Mean $\pm$ std	66.48 $\pm$ 0.06

The 0.06 pp standard deviation across seeds confirms that RF classification is deterministic to within rounding noise at this sample size, and reported accuracy differences across experimental conditions (e.g., feature ablation, cross-model transfer) are not attributable to classifier randomness.

B.30 Binary Detector Baselines for Ordinal Classification

We evaluate whether existing binary AI-text detectors and membership inference methods, when repurposed for ordinal classification, can match our 19-feature RF pipeline.

Binary detectors (Pythia nucleus). Binoculars (Hans et al., 2024) and Fast-DetectGPT (Bao et al., 2024) scores are computed on the Pythia-1.4B nucleus (p=0.95) subset and fed to: (a) a log-likelihood rank classifier (Log-Rank, 64.0%), (b) a combined RF trained on both detector scores (Combined, 74.2%), and (c) the standard 19-feature RF (78.7%). The 4.5 pp gap (74.2% vs. 78.7%) confirms that perplexity distributional features capture ordinal depth structure beyond what generic detection scores provide. Note that Fast-DetectGPT exhibits numerical overflow on Pythia nucleus d1 (mean perturbation score >28,000); results for this cell should be interpreted cautiously. Across all 9 atlas cells, Binoculars and Fast-DetectGPT ordinal accuracy averages ∼35% and ∼32%, respectively.

Membership inference baselines (all 9 cells). DetectGPT (Mitchell et al., 2023) (gpt2-large scoring model, t5-large perturbation model, 5 perturbations) achieves 30.7% mean 4-class accuracy (d1–d3 BalAcc: 24.3%); Table 3 reports 28.8% under the best-of-3 calibration protocol, which selects the optimal threshold configuration per cell. Min-K% Prob (Shi et al., 2024) (K=20) with CORAL ordinal calibration reaches 33.2% (d1–d3 BalAcc: 40.4%); the best single cell is gpt2xl_temp09 at 50.6%. Min-K%'s advantage over DetectGPT (+2.5 pp) is consistent with token-level log-probabilities containing more depth-relevant information than perturbation-based curvature, but both remain within 10 pp of chance. CORAL calibration improves ordinal structure (d1–d3 BalAcc 40.4% vs. 33.3% random) without closing the gap to RF (71.6%), confirming that the bottleneck is the input signal rather than the calibration strategy. Perturbation-based and membership-inference methods (28–33%) confirm that token-level detection scores lack ordinal depth resolution. Cross-perplexity (Binoculars) is a notable exception: when evaluated across all 9 atlas cells with threshold calibration, Binoculars achieves 69.8%; with multi-feature RF calibration (11 cross-perplexity statistics), it reaches 78.1%, matching our 19-feature pipeline within 0.3 pp (§5.1).

Augmented feature set. To test whether detection scores carry complementary ordinal signal, we add Min-K% probability (K=20) and DetectGPT perturbation discrepancy as additional RF features. The 21-feature model (19 + Min-K% + DetectGPT) achieves 78.90% mean 4-class accuracy (+0.48 pp) and 72.20% d1–d3 balanced accuracy (+0.60 pp). Min-K% ranks #3 by feature importance (0.095), behind `p90_ppl` (0.161) and `var_surprisal` (0.126); DetectGPT ranks #17 (0.026). The marginal improvement is consistent across 8 of 9 atlas cells, indicating that Min-K%'s token-level log-probability signal provides weak but real complementary information to the passage-level distributional features. The top-2 features (`p90_ppl` + `var_surprisal`) retain combined importance of 28.7%, confirming that perplexity tail statistics remain the dominant depth markers even after augmentation.

Pairwise AUC and ordinal collapse. Table 30 reports pairwise binary AUC for three nucleus cells, contrasting our 19-feature RF with LR-calibrated detector scores and their combination. Detectors achieve reasonable d0-vs-d1 separation (Combined AUC $\geq$ 0.949), but d1-vs-d2 and d2-vs-d3 AUC collapse toward chance (0.50–0.77), far below the RF pipeline (0.930–0.958 and 0.813–0.892 respectively). This confirms that binary detection scores capture the real-vs-synthetic boundary but lack the ordinal gradient needed to distinguish recursive depths.

Table 30: Pairwise binary AUC for detector baselines vs. RF-19feat on three nucleus cells. RF uses 500-tree binary classifiers; detector methods use LR on continuous scores (Bino/FDGPT) or their combination. d1v2 and d2v3 AUC near 0.50 = ordinal collapse.

Cell	Method	d0v1	d1v2	d2v3
Pythia-1.4B	RF-19feat	**0.999**	**0.930**	**0.813**
	Binoculars	0.738	0.529	0.515
	Fast-DetectGPT	0.797	0.596	0.530
	Combined	0.990	0.767	0.626
OLMo-1B	RF-19feat	**0.997**	**0.934**	**0.819**
	Binoculars	0.744	0.509	0.522
	Fast-DetectGPT	0.917	0.668	0.520
	Combined	0.949	0.742	0.550
GPT-2 XL	RF-19feat	**0.999**	**0.958**	**0.892**
	Binoculars	0.942	0.502	0.581
	Fast-DetectGPT	0.955	0.644	0.655
	Combined	0.969	0.724	0.703

B.31 Binoculars Feature Ablation and Signal Decomposition

To quantify the overlap and complementarity between our 19-feature pipeline and Binoculars' cross-perplexity features, we evaluate four feature configurations across all 9 atlas cells (Table 31).

Table 31: Feature ablation: overlap and merged configurations (9-cell mean, 5-fold CV, RF 200 trees). The 6 overlap features are the perplexity statistics shared between both pipelines (mean, variance, median, p10, p90, skewness).

Configuration	#Feat.	Acc (%)	Δ vs. 19-feat
6 overlap only	6	75.9	−2.5
Our 19 features	19	78.4	—
Binoculars 11 features	11	78.1	−0.3
Merged (19 + 11)	30	**82.6**	+4.2

The 6 shared perplexity features alone achieve 75.9% (97% of the full pipeline's accuracy), confirming that mean-level perplexity distributional displacement is the core depth signal. Our 13 additional features (kurtosis, surprisal variance, lexical/n-gram statistics) contribute +2.5 pp of shape-level information; Binoculars' 5 unique features (observer-performer agreement score, cross-entropy statistics) contribute +4.2 pp of cross-model divergence when merged.

Table 32 reports per-cell results for three configurations with per-cell breakdown (Bino-11 per-cell values are omitted because features were computed on the pooled 9-cell corpus).

Table 32: Per-cell feature ablation (4-class accuracy %, 5-fold CV, RF 200 trees). Δ: merged minus 19-feature accuracy. Bino-11 per-cell values omitted because Binoculars features were computed on the pooled 9-cell corpus; the per-cell mean (78.1%) is reported in Table 31.

Cell	6-overlap	19-feat	Bino-11	Merged	Δ
Pythia nucleus	76.7	78.7	—	83.5	+4.8
Pythia temp09	73.9	76.8	—	81.5	+4.7
Pythia top-k50	65.3	68.9	—	71.2	+2.3
OLMo nucleus	76.4	78.9	—	84.2	+5.3
OLMo temp09	81.0	82.3	—	88.3	+6.0
OLMo top-k50	72.5	74.4	—	78.4	+4.0
GPT-2 XL nucleus	81.7	84.3	—	88.8	+4.5
GPT-2 XL temp09	84.0	85.9	—	88.5	+2.6
GPT-2 XL top-k50	72.1	75.8	—	79.1	+3.3
9-cell mean	**75.9**	**78.4**	**78.1**	**82.6**	**+4.2**

Strategy-dependent complementarity. The merged improvement varies by decoding strategy: nucleus and temperature cells gain +4.7 pp on average, while top-k cells gain +3.2 pp. Top-k sampling truncates the token distribution to the k highest-probability tokens, producing text with reduced distributional diversity; this limits the information available to cross-model divergence features, which rely on disagreement between observer and performer predictions. Merged improvement also varies by generator family: OLMo cells (observer-dissimilar) gain +5.1 pp on average, compared to +3.9 pp for Pythia (observer-similar) and +3.5 pp for GPT-2 XL, suggesting cross-model divergence features contribute more when observer and generator architectures differ.

Cross-scale Binoculars ablation (14B). We evaluate Binoculars cross-scale on two representative strategies (temperature, top-k); the nucleus cell was used for the main 14B pipeline and its GPU hours were allocated to generation rather than Binoculars evaluation. At 14B, when only 1–1.5B reference models are available for cross-perplexity computation (Pythia-1.4B observer analyzing Qwen2.5-14B text), Binoculars features degrade substantially (Table 33). Self-reference statistical features (19-feat) outperform Binoculars (11-feat) by 8–11 pp, reversing the near-parity observed at 1–1.5B. Merging all 30 features yields a consistent +3.6–3.7 pp gain over the self-reference baseline across both tested strategies, confirming that cross-perplexity features retain complementary signal even under the 10× scale gap, though the gain is attenuated relative to +4.2 pp at 1–1.5B.

Table 33: Feature ablation at 14B (Qwen2.5-14B, self-reference†). Binoculars: observer=Pythia-1.4B, performer=Pythia-410M. Δ: merged minus main-pipeline stat-only accuracy (Method A alignment). Chance = 25%.

Strategy	Bino-11	Stat-19†	Merged-30	Δ
temp09	42.6	50.5	54.2	+3.7
top-k50	37.7	47.9	51.5	+3.6

The Binoculars degradation reflects a practical deployment constraint: the 10× scale gap between observer (1.4B) and generator (14B) causes cross-perplexity features to lose sensitivity, not a fundamental limitation of the cross-model divergence channel. Self-reference features are more robust across scales because they avoid the observer–generator mismatch. The accuracy difference between self-reference (50.8%) and Binoculars stat-only (40.1% mean) may therefore partially reflect observer capacity limitations rather than feature design differences.

B.32 Prompted Generation Results

Table 34 reports depth estimation performance under topic-seeded (prompted) generation using GPT-2 XL with nucleus sampling (p=0.95, depths 0–3). Generation is conditioned on Wikipedia topic prompts rather than a bare BOS token, a more ecologically valid test of depth signal persistence.

Table 34: Prompted generation (GPT-2 XL, nucleus p=0.95, topic-seeded). BOS baseline re-evaluated within the prompted-generation pipeline for matched comparison. 5-fold CV accuracy and pairwise AUC. Chance = 25%.

Metric	Value
4-class accuracy (%)	69.52 [68.9, 70.2]
BOS-seeded baseline (%)	83.3
Δ (prompted − BOS)	−13.78 pp
AUC d0v1	0.916
AUC d1v2	0.782
AUC d2v3	0.744
Top-1 feature	`p90_ppl`
Top-2 feature	`var_surprisal`
Top-3 feature	`p75_ppl`

The 13.8 pp drop relative to BOS-seeded generation is expected: topic prompts constrain the output distribution, reducing the variance that distinguishes depths. Nevertheless, accuracy remains 2.8× chance and pairwise AUC decays monotonically (0.916, 0.782, 0.744); the depth signal is therefore not an artifact of unconstrained generation. Top features are identical to BOS-seeded (`p90_ppl` first); the same perplexity tail erosion mechanism operates under prompted conditions.

B.33 Extended Discussion

Cross-strategy conditional decomposition. The two-stage pipeline (strategy identification followed by depth classification) recovers 11.5 pp over the naïve pooled classifier. Under correct strategy routing, four-class accuracy reaches 72.6% vs. 68.0% under misrouting (N=103,498 / 76,502), a gap of 4.6 pp; d0 accuracy remains near-ceiling (∼99.6%) regardless of routing, confirming that the d0 signal is strategy-invariant. Even the worst routing yields a floor of 67.6% (2.7× chance), and the 31.5 pp d1–d3 gap between correct and wrong routing confirms feature-space misalignment rather than uniform noise. End-to-end PLMs lag RF by ∼20 pp, suggesting that current pretrained representations do not capture the distributional tail statistics that drive depth discrimination.

Depth-graduated curation (inconclusive). Initial curation experiments remain inconclusive: RF-vs-unfiltered is not significant ($p > 0.4$; Appendix B.16).

Toward a provenance toolkit. Depth estimation complements existing provenance methods along orthogonal axes. Watermarking (Kirchenbauer et al., 2023) provides high-confidence attribution when the generator cooperates but is inapplicable to unwatermarked text. Source attribution (Uchendu et al., 2020) identifies the generating model but not the generation history. Depth estimation characterizes *iterative reuse* regardless of generator cooperation or identity. A layered provenance system combining these three signals (watermark presence, source model, and recursive generation depth) could support more nuanced data auditing than any single method.

B.34 Computational Resources and Reproducibility

Hardware. All experiments were conducted on 2× NVIDIA RTX PRO 6000 (48 GB) GPUs locally and 2× NVIDIA RTX PRO 6000 (48 GB) on cloud instances. Total computation is approximately 200 GPU·hours, covering the 3 × 3 generation matrix (LoRA fine-tuning + sampling at each depth), feature extraction, depth extension to d5, 7–14B cross-scale experiments, and all ablations.

Software. Key library versions: PyTorch 2.10.0+cu128, Transformers 5.8.1, PEFT 0.7.1, scikit-learn 1.8.0. Additional dependencies: Accelerate 1.13.0, Datasets 4.8.4, XGBoost 3.2.0, NumPy 2.2.6 (Python 3.12.3, CUDA 12.8).

Hyperparameters. LoRA hyperparameters (r=16, α=32, dropout 0.05, lr 2×10^{-4}, batch size 8, 3 epochs) follow Hu et al. (2022). The Random Forest classifier uses 200 trees, Gini criterion, and no max depth, without hyperparameter search; all reported results use 5-fold stratified cross-validation.

Inference cost. Feature extraction requires a single forward pass of the reference model. On a single GPU this takes approximately 2 minutes for 5,000 passages; a CPU-only alternative requires approximately 50 minutes on server-grade hardware. RF classification adds negligible overhead (<1 s).

C Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.